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Luminescence–structure relationships in solids doped with Bi^{3+}†

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The luminescence properties of Bi^{3+} in 177 inorganic compounds are reviewed and rationalized based on two simple and easy-to-use models. These models are applicable to any kind of compound whose crystal structure is known. The calculation of the nephelauxetic function, *he*, involved in one of the models is described step-by-step for an easy implementation in a home-made calculator or machine-learning algorithm. Finally, a criterion based on the experimental value of the Stokes shift is introduced to help in fixing the origin of the Bi^{3+}-related emission.

1. Introduction

Although investigated for decades[1,2] the spectroscopy of Bi^{3+} is increasingly attracting attention, as recent reviews and papers demonstrate.[3–15] However, conflictual spectral assignments persist in the archival literature, mostly because there is a lack for a rational and simple method based on luminescence–structure relationships to fix the origin of the absorption and emission of Bi^{3+} in solids. This is the purpose of the present report to fill this need.

Trivalent bismuth has a $6s^2$ electron configuration in its ground state. Depending on the nature of its chemical surrounding, this ion glows from the UV to the red with Stokes shifts ranging from $\approx$0.25 to $\approx$2.5 eV, quenching temperatures of $\approx$100 K to $\approx$460 K and emission spectral widths from $\approx$0.25 to $\approx$0.9 eV. In some cases, more than one Bi-related emission is observed although Bi^{3+} occupies only one site in the crystal lattice. Attempts to rationalize these behaviors were made through different semi-empirical models involving the nephelauxetic effect of the ligands[16–21] the chemical shift associated with the host lattice,[6] the electronegativity of the nearby cations,[3,4] clustering effects[6,7,22–33] or the formation of an exciton trapped at the Bi site (Bi-TE) as the consequence of an interaction between the electronic states of Bi^{3+} and the electronic states belonging to the crystal lattice.[4,12,34–46] This exciton is perceived a hole trapped at the Bi^{3+} site (forming formally a Bi^{4+} ion) and an electron shared among the nearby metal acceptors. First-principles-based methods applied to the Bi^{3+} spectroscopy constitute very powerful emerging tools, but their computation is time-consuming and still limited to experts.[47–52]

In the present report, we will briefly review the semi-empirical models and introduce a new method to estimate the energy of the ${}^1S_0(6s^2) \rightarrow {}^3P_1(6s^1p^1)$ electron transition of Bi^{3+} in solids, inspired from the model of Duffy and Ingram[16,19] and recent work.[53,54] This method involves the calculation of the nephelauxetic function, *he*, that accounts for the effect of a crystal site on a given luminescent activator. Since the calculation of *he* is not straightforward, we will describe it step by step in a comprehensive and immediately usable manner. This methodology will illustratively be applied to calcium halides and chalcogenides and to a few crystal lattices of optical interest such as CaF_2, YPO_4, Y_2O_3, $Y_3Al_5O_{12}$, Cs_2NaYX_6 (X = halogen),

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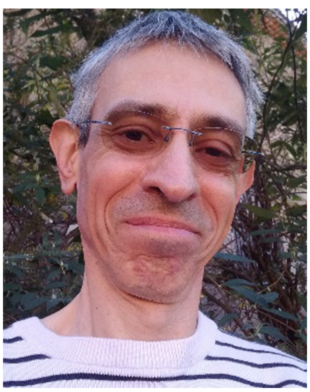
† Electronic supplementary information (ESI) available: The handmade step-by-step calculation of *he* in CaF_2, YPO_4, Y_2O_3, $Y_3Al_5O_{12}$, Cs_2NaYX_6 (X = Cl, Br), $YAlO_3$ and $NaYF_4$. See DOI: https://doi.org/10.1039/d3cp00289f



Philippe Boutinaud

Philippe Boutinaud received his PhD degree in Sept. 1991 from the University of Bordeaux, France, after 2 years spent in the laboratory of Paul Hagenmüller; then he moved to the Department of chemical physics of the University of Geneva, Switzerland, as a research assistant. He got a permanent position as an associate professor in Sept. 1994 at the Laboratory of Inorganic Materials, University of Clermont-Ferrand, France, received his habilitation in Feb. 1997 and was promoted as full professor two years later. His research activity focuses on luminescent materials with specific interest on structure–luminescence relationships and metal-to-metal charge transfer processes.

$YAlO_3$, $NaYF_4$ and will be extended to 177 inorganic compounds. Combined with the metal-to-metal charge transfer model that was established a few years ago,[4] this method enables to: (1) identify the nature of the crystal site that is occupied by Bi^{3+} ion in inorganic solids, (2) fix the origin of the absorption and (3) fix the origin of the emission. For this, only the crystal structure of the compound serves as input.

2. The structure of Bi^{3+} excited states: brief overview of existing models

The excited states of Bi^{3+} have either intra-ionic or extra-ionic character (Fig. 1). In the first situation, we deal with interconfigurational $6s^2 \rightarrow 6s^16p^1$ transitions between the $^1S_0$ ground state and the $^3P_{0-2}$, $^1P_1$ excited states. The $^1S_0 \rightarrow {}^1P_1$ transition is spin allowed and gives rise to intense absorption (labelled C band) usually located in the UV region. The $^1S_0 \rightarrow {}^3P_0$ and $^1S_0 \rightarrow {}^3P_2$ transitions are spin forbidden. The former has negligible absorption cross-section while the latter (labelled B band) is sometimes observed due to coupling with unsymmetrical lattice vibration modes. The spin-orbit coupling between $^1P_1$ and $^3P_1$ contributes to partially allow the $^1S_0 \rightarrow {}^3P_1$ transition (labelled A band), which therefore gains an appreciable oscillator strength. $^3P_1$ and $^3P_0$ are the lowest-lying excited states separated by ≈ 0.002 to ≈ 0.15 eV (*i.e.* the optical trap depth), depending on the considered oxidic host lattice.[11]

The emission from the $^3P_0$ state is observable at low temperature with long lifetime, typically up to few ms. $^3P_1$ is thermally populated upon warming and emits with a much shorter lifetime. It is considered as the principal emitting state at room temperature after internal relaxation (Fig. 1). The corresponding band-like emission is usually referred to as an A emission and $^3P_1$ is thereby usually referred to as the A state. The A emission is in general characterized by relatively small spectral bandwidths and Stokes shifts, although this is not a feature that can be generalized. The energy of the A state is 9.41 eV in the free ion,[55] which locates the corresponding absorption far in the UV. The incorporation of Bi^{3+} in the solid

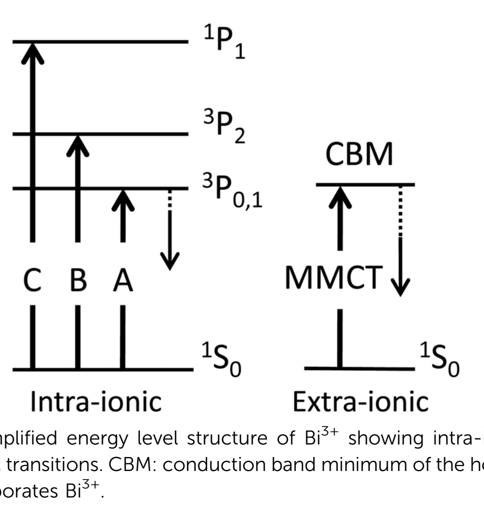


Fig. 1 Simplified energy level structure of Bi^{3+} showing intra-ionic and extra-ionic transitions. CBM: conduction band minimum of the host lattice that incorporates Bi^{3+}.

causes an important energy downshift of this state. This was formalized by Duffy and Ingram in the 70s' though the equation:

$$E_A = E_f(1 - mh) \tag{1}$$

that connects the energy of the A transition (E_A) with the nephelauxetic ratio h of the host lattice.[16,19] In this equation, E_f is the energy of the A transition in correspondence with $h = 0$, *i.e.* the free ion. h expresses the ability for a particular ligand to bring about orbital expansion of the central ion and m is a measure of the extent to which the s and p orbitals of the central ns^2 ion are expanded by different ligands. The values of h amount to 2.0, 2.3 and 2.7 in chlorides, bromides and iodides respectively. These values macroscopically reproduce the trends along the nephelauxetic sequence F, Cl, Br, I in ns^2 halides and ns^2-doped alkali halides[56] and yield a m value of 0.28 for Bi^{3+}. In oxidic media, the m value of Bi^{3+} is lowered to 0.21 and the energy E_f in eqn (1) is found 2.47 eV below the tabulated free ion value. The agreement with experimental data is not as convincing as in halides.[19,57]

Another methodology was introduced in 2012 by L. Wang *et al.*[21] It allows an estimation of the energy of the A and C excitation bands of Bi^{3+} in any kind of lattice, subject that its crystal structure is known. The method involves the calculation of the nephelauxetic function *he*, also referred to as the environmental factor in the paper. The physics behind this concept was formalized by Phillips, Van Vechten and Levine at the end of the 60s'–early 70s' and applied with success in many fields.[58–62] The *he* function accounts for the nephelauxetic effect experienced by a given crystal position M in the lattice. It is defined as:

$$he(\mathrm{M}) = \sqrt{\sum f_c(\mathrm{M{-}Xi})\alpha(\mathrm{M{-}Xi})Q(\mathrm{Xi})^2} \tag{2}$$

where f_c(M–Xi) and α(M–Xi) represent respectively the fractional covalency and the volume polarization of each chemical bond separating cation M to its nearest ligands Xi in binary units to which the host lattice must initially be decomposed. Q(Xi) is the charge of the anions. The sum is extended to the coordination number of crystal site M. The knowledge of he(M) was notably used to reproduce the centroid energy of $4f^{n-1}5d^1$ configurations of Ce^{3+} and Eu^{2+} in solids[63] and to formulate a relationship between the host lattice and sp energy levels of Bi^{3+}, Pb^{2+} and Sb^{3+} ions.[5,12,53,54,64,65]

The energy $E_{sp}(Bi^{3+})$ associated with the $s^2 \rightarrow$ sp transitions of Bi^{3+} ions located in the crystal position M of a lattice was expressed in eV as:[21]

$$E_{sp}\left(Bi^{3+}\right) = 2.972 + 6.206\exp\left(-\frac{he(\mathrm{M})}{0.551}\right) \tag{3}$$

This model reproduces the experimental data within $\approx \pm 0.5$ eV but does not reproduce the free ion energy accurately (9.18 eV against 9.41 eV expected).[21]

More recently, Awater and Dorenbos have reviewed the optical properties of 117 compounds doped with Bi^{3+} and demonstrating a linear increase of the A excitation energy as a function of the Coulomb repulsion energy of the host lattice, U.[6]

The accuracy of method amounts to $\approx \pm 0.5$ eV and, unlike the previous method, does not give information on the crystal site occupied by Bi^{3+}.

When the host lattice is a d^0 or d^{10} closed-shell complex metal oxide (*i.e.* titanates, zirconates, vanadates, niobates, tantalates, molybdates, tungstates, gallates, stannates, *etc.*), a trapped exciton (TE) can form as the consequence of an electron transition from the $^1S_0$ ground state of Bi^{3+} to the conduction band minimum (CBM) of the host lattice (Fig. 1).[4,12,34–46] This extra-ionic transition (also called D transition) is also described as resulting from a metal-to-metal charge transfer (MMCT) of the type $s^2 + d^0 \rightarrow s^1 + d^1$ or $s^2 + d^{10}s^0 \rightarrow s^1 + d^{10}s^1$, where Bi^{3+} is the electron donor and the adjacent d^0 or d^{10} metal cation is the electron acceptor. Depending on the electronegativity of the metal acceptor and its distance from Bi^{3+}, the corresponding CT occurs at higher, equivalent or lower energy relative to the intra-ionic A transition. Fig. 2 shows the electronegativities of a few d^0 and d^{10} metal ions adopting either an octahedral (CN′ = 6) or tetrahedral (CN′ = 4) coordination in solids.[66]

Redox processes can also occur between two nearby Bi^{3+} ions if the Bi^{3+} content is high enough in the compound. In the latter case, the Bi → Bi CT is identified as an intervalence charge transfer (IVCT) of the type $Bi^{3+}(6s^2) + Bi^{3+}(6s^2) \rightarrow Bi^{4+}(6s^1) + Bi^{2+}(6s^2p^1)$.[6] MMCT and IVCT processes involve an outer-sphere charge transfer in which the orbital overlap between the involved atoms is expected small or inexistent. In this scheme, the electron donating/accepting capacity and the distance between the atoms at their closest possible approach critically affect the value of the charge transfer energy; the energy of the outer-sphere CT transition becoming larger upon increasing the donor–acceptor distance. This is reproduced by the equation:[4]

$$E_{\mathrm{CT}}(\mathrm{Bi}^{3+}) = k_{\mathrm{CN}'}^{\mathrm{CN}}\left[\chi_{\mathrm{CN}}(\mathrm{Bi}^{3+}) - \alpha_{\mathrm{CN}'}^{\mathrm{CN}}\frac{\chi_{\mathrm{CN}'}(\mathrm{acceptor})}{d_{\mathrm{Bi\text{-}acceptor}}}\right] \quad (4)$$

where $d_{\mathrm{Bi\text{-}acceptor}}$ is the distance separating the Bi^{3+} donor with coordination number CN to its nearest metal acceptor with coordination number CN′ (*viz.* the d^0 or d^{10} metal in case of MMCT or Bi^{3+} in case of IVCT), corrected for ionic radii mismatches between Bi^{3+} and the substituted cation site M. This mismatch creates a local distortion at the doping site that alters the M–Xi and Bi-acceptor distances according to $d_{\mathrm{Bi–Xi}} = d_{\mathrm{M–Xi}} + \Delta d$ or $d_{\mathrm{Bi\text{-}acceptor}} = d_{\mathrm{M\text{-}acceptor}} + \Delta d$, where $\Delta d = \frac{1}{2}[r(\mathrm{Bi}) - r(\mathrm{M})]$ with $r(\mathrm{Bi})$ and $r(\mathrm{M})$ the ionic radii of the considered cations as picked up from ref. 67. Indicative values of Δd are given in Table 1.

$\chi_{\mathrm{CN}}(\mathrm{Bi}^{3+})$ is the electronegativity of Bi^{3+} in its doping site and $\chi_{\mathrm{CN}'}(\mathrm{acceptor})$ is the electronegativity of the nearby d^0 or d^{10} metal cation (or Bi^{3+} in pairs). All data are picked up from ref. 66 or from Fig. 2. $k_{\mathrm{CN}'}^{\mathrm{CN}}$ and $\alpha_{\mathrm{CN}'}^{\mathrm{CN}}$ are crystal dependent adjustable parameter whose values are compiled in Table 2. The accuracy of the method amounts to ± 0.35 eV.[4,8]

The above semi-empirical models allow to estimate the energy of the absorbing states but do not predict the origin of the Bi^{3+}-related emission. Dealing with isolated Bi^{3+} ions, three situations may occur depending on the energy separating A and MMCT states, or, in other words, on the location of the A state relative to the CBM. These three situations are depicted in Fig. 3. In situation (a), the A state is located well below the CBM. If not excited, the MMCT state does not interfere with the relaxation process and the spectroscopy of Bi^{3+} is driven by intra-ionic transitions. This concerns host lattices with large bandgaps. The emission usually appears in the UV or blue spectral region. If the MMCT state is excited, then it can either emit or relax to the lower-lying A state. In situation (c), the A state is located within the host conduction band. The lowest excitation of Bi^{3+} reaches the CBM and forms a trapped exciton that emits. The emission appears in the visible, usually with larger Stokes shifts and larger spectral bandwidths, revealing a stronger electron–phonon coupling. Situation (b) is more complex. Here, the A and MMCT states are close to each other, and emission can occur from both states. This depends on the energy mismatch between the states and on their relative positions (*i.e.* A located below or above the CBM). The relaxation process is, in this case strongly temperature dependent.

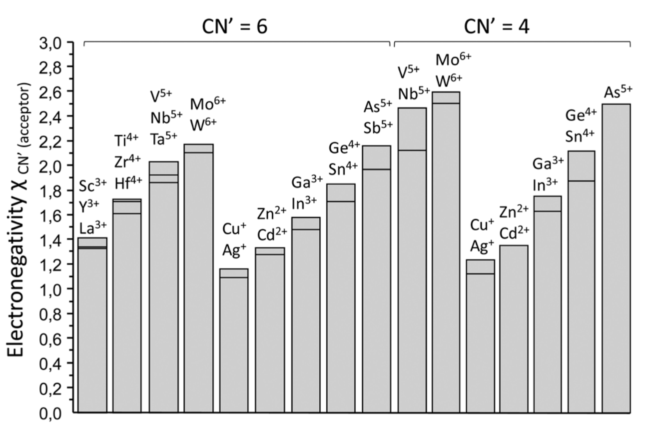


Fig. 2 Electronegativities of d^0 and d^{10} metal cations in octahedral (CN′ = 6) or tetrahedral (CN′ = 4) coordination.[66]

Table 1 Local correction Δd after Bi^{3+} substitution. Values in Å

M	Cs^+	Rb^+	K^+	Ba^+	Pb^{2+}	Sr^{2+}	Na^+	Ca^{2+}	La^{3+}	Cd^{2+}	Gd^{3+}
Δd	−0.285	−0.245	−0.175	−0.125	−0.07	−0.05	0			+0.035	+0.06

M	Y^{3+}	Lu^{3+}	In^{3+}	Li^+	Zn^{2+}	Mg^{2+}	Sc^{3+}	Sn^{4+}	Ga^{3+}
Δd	+0.075	+0.09	+0.12	+0.135		+0.15		+0.175	+0.20

Table 2 Value of parameters $k^{CN}_{CN'}$ and $\alpha^{CN}_{CN'}$ in eqn (4) for MMCT[4] and IVCT[8] transitions as a function of CN and CN′

		MMCT transitions				IVCT transitions	
CN	$\chi_{CN}(Bi^{3+})$	$k^{CN}_{CN'=4}$ (eV)	$\alpha^{CN}_{CN'=4}$	$k^{CN}_{CN'\geq 6}$ (eV)	$\alpha^{CN}_{CN'\geq 6}$	$k^{CN}_{CN'}$ (eV)	$\alpha^{CN}_{CN'}$
5	1.43	6.07	1.06	4.77	1.18	8.00	2.58
6	1.40	6.20	1.04	4.87	1.16	8.16	2.52
7	1.37	6.33	1.02	4.98	1.13	8.34	2.46
8	1.34	6.48	0.99	5.09	1.11	8.53	2.41
9	1.33	6.52	0.99	5.13	1.10	8.59	2.39
10	1.305	6.65	0.97	5.22	1.08	8.76	2.35
12	1.28	6.78	0.95	5.33	1.06	8.93	2.30

The origin of Bi^{3+} emission is solids has been discussed for long based on spectral bandwidths and Stokes shifts considerations. These arguments are usually valid but cannot be generalized. To cover all possible situations, a criterion was introduced in ref. 12 and 14. It consists of calculating the absolute value of the energy difference:

$$\Delta E(\mathrm{Bi}^{3+}) = |E_{\mathrm{CT}}(\mathrm{Bi}^{3+}) - E_{\mathrm{A}}(\mathrm{Bi}^{3+}) - S| \quad (5)$$

where $E_{CT}(Bi^{3+})$ and $E_A(Bi^{3+})$ are calculated positions of CT and A states. $E_{CT}(Bi^{3+})$ is calculated by means of eqn (4). In the present work, $E_A(Bi^{3+})$ will be calculated using the method introduced in Section 4. S corresponds to the experimental Stokes shift associated with the bismuth luminescence in the host lattice of interest, *i.e.* the energy separating the excitation and emission bands maxima. Note that these values do not necessarily correspond to "true" Stokes shifts in the sense that the absorbing state is not necessarily the emitting state. This is exemplified in Fig. 3.

The basic idea of eqn (5) is that if $E_{CT}(Bi^{3+}) > E_A(Bi^{3+})$, then the CT state is located above the A state and interferes little with the emission process (situation (a) in Fig. 3). Then, subtracting S to $E_{CT}(Bi^{3+})–E_A(Bi^{3+})$ contributes to lower $\Delta E(Bi^{3+})$; the lower the absolute value of this amount, the more probable the regular A emission for Bi^{3+}. Conversely, if $E_{CT}(Bi^{3+}) < E_A(Bi^{3+})$, then the CT state is located below the A state (situation (c) in Fig. 3) and may impact the emission process deeply by creating an additional relaxation path leading to CT emission. Then, subtracting S to $E_{CT}(Bi^{3+})–E_A(Bi^{3+})$ leads to more negative value of $\Delta E(Bi^{3+})$ that finally gives a large absolute value, the larger the amount, the more probable the CT emission for the compound. Situations where $E_{CT}(Bi^{3+}) \approx E_A(Bi^{3+})$, *i.e.* situation (b) in Fig. 3, give $\Delta E(Bi^{3+}) \approx S$. In this case, the larger S, the more probable a CT emission. It was reported in ref. 12 that the Bi^{3+} emission is of CT character when $\Delta E(Bi^{3+}) > 1$ eV and of A character otherwise. This criterion will be refined at the light of the new results presented in this report.

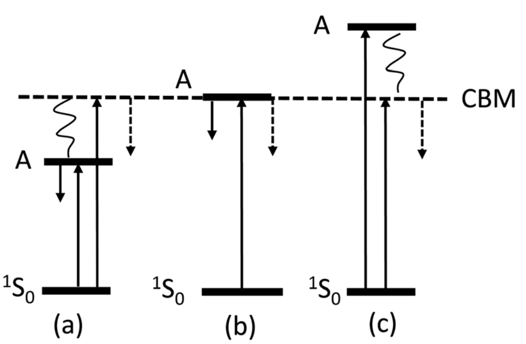


Fig. 3 Comparative scheme showing the possible relative positions of the A state relative to the CBM of the host. The descending arrow in solid line pertains to an emission from the A state while the arrow in dotted line pertains to an emission from the MMCT state.

3. The step-by-step calculation of *he*(M)

In this section, we propose a step-by-step method for an easy calculation of *he*(M) in a compound with general formula $A_aM_mX_x$, where A and M are cations and X are anions. In this compound, we will consider M as the doping site for the luminescent activator. We will concentrate only on the algebraic procedures that are involved in the calculation of *he*(M). The physics behind the theory is let aside.

If cations and anions occupy different crystal sites $i = 1, 2, \ldots, n$ in $A_aM_mX_x$, then the compound must firstly be written as $\mathrm{Ai}_{ai}\mathrm{Mi}_{mi}\mathrm{Xi}_{xi}$, where $\sum_{i=1}^{n} a_i = a$, $\sum_{i=1}^{n} m_i = m$, $\sum_{i=1}^{n} x_i = x$. The coefficients a_i, m_i and x_i are obtained as $a_i = w(\mathrm{Ai})/Z$, $m_i = w(\mathrm{Mi})/Z$, $x_i = w(\mathrm{Xi})/Z$, where w(Ai), w(Mi) and w(Xi) are the multiplicity of the different Wyckoff sites occupied by atoms Ai, Mi and Xi in the crystal lattice and Z is the number of formula units per unit cell. In the present Section, we will deal with single cation sites A1 and M1 (that we will simply write as A and M in the following) and with two anion sites X1 and X2. Our compound is then $A_aM_mX1_{x1}X2_{x2}$. This corresponds for instance to the situation in the orthorhombic perovskite $YAlO_3$.

3.1. Input structural data and crystal decomposition

The first step consists in decomposing $A_aM_mX1_{x1}X2_{x2}$ into a sum of binary units made of cations A and M surrounded by anions X1 and X2, following:

$$\mathrm{A}_a\mathrm{M}_m\mathrm{X1}_{x1}\mathrm{X2}_{x2} = \mathrm{A}_{K(\mathrm{A-X1})}\mathrm{X1}_{K(\mathrm{X1-A})} + \mathrm{A}_{K(\mathrm{A-X2})}\mathrm{X2}_{K(\mathrm{X2-A})} + \mathrm{M}_{K(\mathrm{M-X1})}\mathrm{X1}_{K(\mathrm{X1-M})} + \mathrm{M}_{K(\mathrm{M-X1})}\mathrm{X2}_{K(\mathrm{X2-M})} \quad (6)$$

With the Kappa function defined as:

$$K(\mathrm{A-Xi}) = \frac{w(\mathrm{A})}{Z}\frac{N(\mathrm{A-Xi})}{N(\mathrm{A})},\ K(\mathrm{Xi-A}) = \frac{w(\mathrm{Xi})}{Z}\frac{N(\mathrm{Xi-A})}{N(\mathrm{Xi})},$$
$$K(\mathrm{M-Xi}) = \frac{w(\mathrm{M})}{Z}\frac{N(\mathrm{M-Xi})}{N(\mathrm{M})} \text{ and } K(\mathrm{Xi-M}) = \frac{w(\mathrm{Xi})}{Z}\frac{N(\mathrm{Xi-M})}{N(\mathrm{Xi})} \quad (7)$$

where $i = 1$ for anion X1 and 2 for anion X2. In these expressions, N(A), N(M) and N(Xi) are the coordination numbers of the considered atoms, N(A–Xi) and N(M–Xi) represent the number of anions Xi that surround cations A and M, respectively, while N(Xi–A) and N(Xi–M) represent the number of cations A and M that surround anions Xi, respectively.

These input structural data are conveniently compiled in the form of Matrix I below. They are easily obtained from CIF files using *e.g.* VESTA.[68] These data have great impact on the final *he* values and must be picked up with care.

Matrix I	A	M	X1	X2
A			N(X1–A)	N(X2–A)
M			N(X1–M)	N(X2–M)
X1	N(A–X1)	N(M–X1)		
X2	N(A–X2)	N(M–X2)		
Total coordination	N(A) = N(A–X1) + N(A–X2)	N(M) = N(M–X1) + N(M–X2)	N(X1) = N(X1–A) + N(X1–M)	N(X2) = N(X2–A) + N(X2–M)
Site multiplicity	w(A)	w(M)	w(X1)	w(X2)

At this stage, we pick up the interatomic distances between cations and anions in the coordination polyhedra and the bond valence parameters (BVP) for all cations in oxides, halides, *etc.* that are retrieved in ref. 69. We compile these data in the form of Matrix II for further use.

Matrix II	A	M
X1	d(A–X1)	d(M–X1)
X2	d(A–X2)	d(M–X2)
Bond valence parameter	BVP(A)	BVP(M)

3.2. Calculation of *he*(M) in $A_aM_mX1_{x1}X2_{x2}$

The application of eqn (2) to site M in $A_aM_mX1_{x1}X2_{x2}$ gives:

$$he(\mathrm{M}) = [N(\mathrm{M-X1})f_c(\mathrm{M-X1})\alpha(\mathrm{M-X1})Q(X1)^2 + N(\mathrm{M-X2})f_c(\mathrm{M-X2})\alpha(\mathrm{M-X2})Q(\mathrm{X2})^2]^{1/2} \quad (8)$$

At this stage, we split the calculation into 3 parts.

3.2.1. Calculation of *Q*(Xi). The values of Q(Xi) are obtained from the effective charge Q(M) carried by cation M in the crystal as:

$$Q(\mathrm{Xi}) = \frac{K(\mathrm{M-Xi})}{K(\mathrm{Xi-M})}Q(\mathrm{M}) \quad (9)$$

The Kappa values are calculated by using eqn (7) and Matrix I. The bond valence sum of atom M is considered as a good approximation of Q(M) This amount is easily calculated from BVP(M) using the facilities implemented in VESTA[68] when all anions Xi have same chemical nature. When the anions are different in a compound, *e.g.* oxygens and chlorides in the example of YOCl, then Q(M) is calculated using the original equation:[68]

$$Q(\mathrm{M}) = \sum_{i=1}^{N(\mathrm{M})} \exp -\frac{d(\mathrm{M-Xi}) - \mathrm{BVP(M)}}{0.37} \quad (10)$$

taking the BVP(M) value that corresponds to the ligands' nature.

Considering the example of YOCl (ICSD 31667), we have 4 Y–O distances at 2.283 Å and 4 Y–Cl distances at 3.002 Å and we find in ref. 69: BVP(Y^{3+}) = 2.014 for oxygen ligands and BVP(Y^{3+}) = 2.40 for chlorine ligands. Following (10), we calculate $Q(\mathrm{Y}) = 4\left[\exp -\frac{2.283 - 2.014}{0.37} + \exp -\frac{3.002 - 2.40}{0.37}\right] = 2.72.$ By the same method, we find Q(Y) = 2.58 for YOF, Q(Gd) = 3.16 for GdOCl, Q(La) = 2.9 for LaOCl, Q(La) = 2.96 for LaOBr, Q(La) = 3.09 for La_2O_2S, Q(Ca) = 1.89 for CaZnOS, Q(Ba) = 1.86 for $BaScO_2F$, Q(Sr1) = 1.36 and Q(Sr2) = 2.17 for $SrAlO_4F$.

Valence state (VS) and valence numbers (VN) of the anions will also be useful in the following. The values are given in Matrix III for different kinds of ligands.

Matrix III	Xi = group VII	Xi = group VI	Xi = group V
VS(Xi)	1	2	3
VN(Xi)	7	6	5

3.2.2. Calculation of f_c(M–Xi). The general expression is as follows:

$$f_c(\mathrm{M-Xi}) = \frac{1}{1 + \frac{[\Gamma_1(\mathrm{M-Xi})\Gamma_2(\mathrm{M-Xi})\Gamma_3(\mathrm{M-Xi})]^2}{1579d(\mathrm{M-Xi})^{-4.96}}} \quad (11)$$

in which:

$$\Gamma_1(\mathrm{M-Xi}) = \frac{2.56\left[\frac{K(\mathrm{M-Xi})N(\mathrm{M}) + K(\mathrm{Xi-M})N(\mathrm{Xi})}{K(\mathrm{M-Xi}) + K(\mathrm{Xi-M})}\right]^2}{d(\mathrm{M-Xi})} \quad (12)$$

$$\Gamma_2(\mathrm{M-Xi}) = Q(\mathrm{M})\left[1 - \frac{\mathrm{VN(Xi)}}{\mathrm{VS(Xi)}}\right] \quad (13)$$

$$\Gamma_3(\mathrm{M-Xi}) = \exp\left[-1.364d(\mathrm{M-Xi})\frac{N^*(\mathrm{M-Xi})^{\frac{1}{6}}}{\upsilon(\mathrm{M-Xi})^{\frac{1}{6}}}\right] \quad (14)$$

with

$$N^*(\mathrm{M-Xi}) = \frac{Q(\mathrm{M})}{N(\mathrm{M})} + \frac{\mathrm{VN(Xi)}}{\mathrm{VS(Xi)}}\frac{K(\mathrm{M-Xi})}{K(\mathrm{Xi-M})}\frac{Q(\mathrm{M})}{N(\mathrm{Xi})} \quad (15)$$

and

$$\upsilon(\mathrm{M-Xi}) = \frac{d(\mathrm{M-Xi})^3}{\sum_i \frac{Zm_iN(\mathrm{M-Xi})}{V}d(\mathrm{M-Xi})^3 + \sum_i \frac{Za_iN(\mathrm{A-Xi})}{V}d(\mathrm{A-Xi})^3} \quad (16)$$

V is the unit cell volume. All other parameters are picked up in Matrices I, II and III or calculated from eqn (7). As stated already, Q(M) is the bond valence sum directly given by VESTA. Again, i = 1 for anion X1 and 2 for anion X2.

3.2.3. Calculation of α(M–Xi)

$$\alpha(\mathrm{M-Xi}) = \frac{0.208 f_c(\mathrm{M-Xi}) d(\mathrm{M-Xi})^{4.96} N^*(\mathrm{M-Xi})}{3 + 0.874 f_c(\mathrm{M-Xi}) d(\mathrm{M-Xi})^{4.96} \dfrac{N^*(\mathrm{M-Xi})}{\upsilon(\mathrm{M-Xi})}} \quad (17)$$

All necessary parameters are picked up in Matrices I, II and III or calculated from eqn (11), (15) and (16).

Finally, *he*(M) can be calculated from eqn (8). In many cases, the calculation of *he*(M) is achievable by hand with some patience. This is illustrated in the ESI† for several representative compounds of interest in the field of spectroscopy: CaF_2, YPO_4, Y_2O_3, $Y_3Al_5O_{12}$, Cs_2NaYX_6 (X = halogen), $YAlO_3$ and $NaYF_4$. In the context of the present work, a *he* calculator was designed which can manage up to 6 different cation sites and 8 different anions sites in a compound. This covers a large variety of inorganic crystal structures, for instance the monoclinic compound $SrGa_2O_4$ (2 Sr sites, 4 Ga sites and 8 oxygen sites). Its robustness was challenged based on >200 repeatedly checked calculations and we consider it now as a reliable tool. Based on this, a few errors contained in our earlier version of the *he* calculator were corrected and several values of *he* reported earlier[5,12] were amended.

It must be kept in mind that the *he* calculator is, in its present form, limited to crystal sites having regular structural attributes as they are described in crystallographic databases. Any structural modifications resulting from charge compensation, vacancy, *etc.* are not yet implemented. In this frame, we found that *he* varies little with respect to the input structural data. To illustrate this aspect, we have tested our calculator on 25 structural descriptions of the orthorhombic (room temperature) form of $CaTiO_3$ (space group no. 62), covering the period 1957–2020. We have also compared the results to the ideal cubic form of $CaTiO_3$. The ideal cubic perovskite (space group no. 221) is characterized by one Ca site with *w*(Ca) = 1 and *N*(Ca–O) = 12, one Ti site with *w*(Ti) = 1 and *N*(Ti–O) = 6 and one O site with *w*(O) = 3, *N*(O–Ca) = 4 and *N*(O–Ti) = 2. The orthorhombic distortion leads to *w*(Ca) = *w*(Ti) = 4 and generates two anions O1 and O2 with *w*(O1) = 4 and *w*(O2) = 8. From the descriptions available in ICSD, we find *N*(Ca–O1) = 2, *N*(Ca–O2) = 6, *N*(Ti–O1) = 2, *N*(Ti–O2) = 4, *N*(O1–Ca) = 2, *N*(O2–Ca) = 3, *N*(O1–Ti) = 2 = *N*(O2–Ti) = 2. The values of *he* at Ca and Ti sites and a few other structural parameters of $CaTiO_3$ are collected in Table 3. We can note the important effect of the coordination number at the Ca site on the *he*(Ca) value. For a fixed coordination number, we find that *he* values are reasonably reproducible, especially using the most recent structural data.

On these grounds, we have applied the *he* calculator to a series of ''simple'' calcium halides and chalcogenides, as detailed in Table 4. We find that the *he*(Ca) values reproduce the sequence F < Cl < Br < I < O < S < Se < Te in these systems.

Table 3 The *he*(Ca) and *he*(Ti) values in $CaTiO_3$ from different structural inputs

ICSD	Shortest Ca–Ti distance (Å)	*N*(Ca)	*Q*(Ca)	*Q*(Ti)	*he*(Ca)	*he*(Ti)
165076	3.397	12	1.36	4.04	0.299	1.596
16688	3.193	8	1.57	4.45	0.791	1.631
19793	3.170		1.91	4.10	0.828	1.587
37263	3.170		1.91	4.10	0.867	1.585
51204	3.170		1.91	4.10	0.829	1.585
62149	3.166		1.91	4.11	0.827	1.585
71915	3.179		1.88	4.09	0.826	1.585
74212	3.179		1.83	4.18	0.820	1.596
85101	3.164		1.93	4.10	0.830	1.584
88347	3.180		1.90	4.07	0.830	1.584
93082	3.166		1.91	4.12	0.827	1.585
94568	3.264		1.85	4.13	0.838	1.596
97828	3.167		2.03	4.04	0.864	1.586
150420	3.170		1.90	4.12	0.827	1.587
153172	3.165		1.93	4.11	0.830	1.585
162908	3.162		1.93	4.16	0.829	1.590
163528	3.172		1.91	4.12	0.830	1.588
163662	3.167		1.92	4.11	0.830	1.586
165801	3.169		1.93	4.11	0.830	1.587
167888	3.167		1.93	4.11	0.830	1.587
174127	3.179		1.90	4.10	0.828	1.586
183209	3.178		1.92	4.12	0.831	1.587
185443	3.166		1.92	4.10	0.829	1.584
186224	3.169		1.89	3.96	0.817	1.571
189135	3.168		1.95	4.07	0.834	1.583
193648	3.169		1.91	4.11	0.829	1.587
Average (standard deviation)					0.830 (0.01)	1.588 (0.01)

Table 4 The *he*(Ca) values in calcium halides and chalcogenides

Compound	ICSD	*N*(Ca)	BVP(Ca)[69]	*Q*(Ca)	*he*(Ca)
CaF_2	656448	8	1.842	1.94	0.29
$CaCl_2$	9009084	6	2.37	2.18	0.90
$CaBr_2$	56763	6	2.49	2.02	0.96
CaI_2	52580	6	2.72	2.01	1.13
CaO	7200686	6	1.967	1.835	1.35
CaS	603165	6	2.45	2.06	1.96
CaSe	41957	6	2.56	2.05	2.10
CaTe	41958	6	2.76	1.96	2.34

4. The excitation of Bi^{3+} in solids: locating the A state from structural data

Recently, the model of Duffy and Ingram was reformulated for Sb^{3+} and Bi^{3+} in a few halides.[53,54] It was found that the energy of the A transition obeys a power law evolution of the form:

$$E_A(Bi^{3+}) = E_f(1 - Khe^{\alpha}) \quad (18)$$

where *K* and α are phenomenological parameters and *he* stands for *he*(M) or *he*(Bi^{3+}). In the present report, we have calculated the *he* and *he*(Bi^{3+}) values for 242 crystal sites related to 177 compounds doped with Bi^{3+} for which the crystal structure is available either from either the International Crystal Structure Database (ICSD-) provided by FIZ Karlsuhe, the Materials Project database (MP-)[70] or directly from original papers. For *he*(Bi^{3+}), the local distortion induced by Bi^{3+} doping was taken into account by using d_{Bi-Xi} instead of d_{M-Xi} in Matrix II, and the necessary values of Δ*d* listed in Table 1. Since the *he* and CT models are sensitive to the structural attributes of the crystal sites, the compounds in Table 5 are identified as ''Crystal lattice (doping site [CN], metal acceptor [CN′])'' where CN is the coordination of the cation site available for Bi^{3+} and CN′ is the coordination number of the d^0 or d^{10} metal acceptor (or Bi^{3+}

Table 5 Structural and optical properties of inorganic compounds doped with Bi^{3+}. $E_{exp}(Bi^{3+})$ is the experimental excitation energy and S is the Stokes shift associated with emission, as picked up from the archival literature. The type (A or CT) of the considered transition follows the archival literature. Underlined data and compounds are reproduced with eqn (18) within ±0.35 eV using $he(Bi^{3+})$ values. Data and compounds in bold are reproduced with eqn (4) within ±0.35 eV. This includes Bi pairs. Compounds and crystal sites in italic are not reproduced by the models. All energies in eV

	Crystal lattice (doping site [CN], metal acceptor [CN′])				$E_{exp}(Bi^{3+})$		S			
Crystal database reference	he(M)	$he(Bi^{3+})$	U	$\chi_{CN'}/d$	A type	**CT type**	A type	**CT type**	Ref.	$\Delta E(Bi^{3+})$
Free ion	0	0			9.41					
Cs_2HfCl_6 (Cs[12], —)	0.19	0.14			*3.54*		0.85		94	
MP-1120741										
$NaYF_4$(Na1/Y1[9],—)	0.20	0.20	7.51		*4.2 (sh)*		1.38		95	
ICSD-51917					*5.0*					
					5.8–6.5					
Cs_2ZrCl_6 (Cs[12], —)	0.21	0.14			*3.49*		0.77		96	
ICSD-26695										
CaF_2(Ca[8],—)	0.29	0.29	7.31		5.86		0.35		97	
ICSD-2300449										
$NaYF_4$ (Y2[9], —)	0.30	0.33	7.51		*4.2 (sh)*		1.38		95	
ICSD-51917					*5.0*					
					5.8–6.5					
$BaScO_2F$ (Ba[12], Sc[6])	0.33	0.28			*2.99*		0.54		98	
ICSD-19622					*3.44*					
					4.22					
$CaBaZn_2Ga_2O_7$(Ba[12],Ga1-2[4])	0.35	0.31		0.49	*3.81*		0.66		99	
ICSD-194164										
SrF_2(Sr[8],—)	0.35	0.32	7.32		5.78		0.31		97	
ICSD-404414										
$NaLaTa_2O_7$ (La[12], Ta[6]) (↑)	0.37	0.37		0.54	**4.13**		1.70		100	**3.22 (CT)**
ICSD-84363										
$BaSc_2O_4$ (Ba1[12], Sc1[6])	0.37	0.34		0.42	*3.91*		1.10		101	
$BaSc_2O_4$ (Ba1[12], Sc2[6])				0.28	**4.63**					
$BaSc_2O_4$ (Ba1[12], Sc3[6])				0.39						
ICSD-35592										
BaF_2(Ba[8],—)	0.40	0.33	7.38		5.75		1.00		97	
ICSD-9009004										
Sr_3AlO_4F (Sr1[10], —)	0.41	0.40	6.8		*3.97*		1.07		102	
ICSD-253796										
$LaAlO_3$ (La[12], La[12]) (↑)	0.42	0.42	6.67	0.32	*4.35*–**4.56**		1.04–1.02		103 and 104	1.29 (A)
ICSD-182612										
$La_2SrSc_2O_7$ (La/Sr1[12], Sc[6])	0.43	0.43		0.40	*3.64*		1.30		105	
ICSD-167047										
$KBaBP_2O_8$(K/Ba[10],—)	0.43	0.37	7.32		5.17		1.97		106	
ICSD-112241										
$KLaTa_2O_7$ (La[12], Ta[6]) (↑)	0.43	0.43		0.54	**3.81**		1.56		100	**2.93 (CT)**
ICSD-83909										
$NaNbO_3$ (Na1[9], Nb[6]) (↑)	0.45	0.45	6.8[a]	0.55		**3.78**		1.61	5	**2.99 (CT)**
ICSD-262695										
$RbLaTa_2O_7$ (La[12], Ta[6]) (↑)	0.48	0.48		0.54	**4.00**		1.70		100	**2.95 (CT)**
ICSD-86208										
Sr_2GdGaO_5 (Sr2[10], Ga [4])	0.49	0.47		0.53	*3.97*		1.34		27	
ICSD-1183										
$GdTa_7O_{19}$ (Gd/Ta1[8], Ta2[6]) (↑)	0.50	0.53	6.7, 6.8[a]	0.51		**4.00**		1.42	107	**2.40 (CT)**
Ref. 107										
NaCl (Na[6],—)	0.54	0.54	6.7		*3.81*				108	
ICSD-9003308										
$BaLa_2ZnO_5$ (Ba[10], La[8])	0.55	0.49	6.55	0.34	*3.87*				109	
ICSD-88598										
$BaSc_2O_4$ (Ba2[10], Sc1[6])	0.61	0.54		0.43	*3.91*		1.10		101	
$BaSc_2O_4$ (Ba2[10], Sc2[6])				0.39	**4.63**					
$BaSc_2O_4$ (Ba2[10], Sc3[6])				0.41						
ICSD-35592										
$NaNbO_3$ (Na2[8], Nb[6]) (↑)	0.62	0.62	6.8[a]	0.56		**3.78**		1.61	5	**2.70 (CT)**
ICSD-262695										
GdOCl (Gd[9], —)	0.67	0.72	6.46		*3.67*		0.88		22	
GdOCl(Gd[9],Bi[9])(pairs)				0.32		**4.82**		2.03		
ICSD-77820										
YOF(Y[8],—)	0.67	0.74	6.7		4.62		0.86		1	
ICSD-28684										
$LiNbO_3$ (Li[6], Nb[6]) (↑)	0.69	0.79	6.8[a]	0.58		**3.59**		1.40	5	**2.32 (CT)**
ICSD-35953										
YOCl(Y[9],—)	0.63	0.70	6.47		4.90		2.09		1, 22 and 110	

Table 5 (*continued*)

	Crystal lattice (doping site [CN], metal acceptor [CN′])									
					E_{exp} (Bi^{3+})		S			
Crystal database reference	he(M)	he(Bi^{3+})	U	$\frac{\chi_{CN'}}{d}$	A type	CT type	A type	CT type	Ref.	ΔE(Bi^{3+})
YOCl(Y[9],Bi[9])(pairs)				0.32		**4.87**		1.87		
ICSD-31667										
KCl (K[6], —)	0.71	0.57	6.8		*3.70–3.80*		0.82–0.96		108 and 111	
ICSD-9003112										
LaOCl(La[9],—)	0.73	0.73	6.44	0.31	4.56–4.59		0.99–1.06		1, 22, 110	
ICSD-40297					4.80		1.33		and 112	
$Li_4SrCa(SiO_4)_2$ (Sr[8], —)	0.75	0.71	6.8		*4.00*		0.74		113	
ICSD-259898										
LaB_3O_6 (La[10],La[10]) (↓)	0.75	0.75	7.15	0.31	**4.77**		1.01		24	0.47 (A)
LaB_3O_6 (La[10],Bi[10])(pairs)				0.32		**4.77**		1.60		
ICSD-415962										
$LaNbO_4$ (La[8], Nb[6]) (↑)	0.76	0.76	6.8	0.51		**3.93**		1.40	5	**1.97 (CT)**
ICSD-73390										
$LiTaO_3$(Li[6],Ta[6])(↑)	0.76	0.84		0.60	4.13		1.22		114	**2.18 (CT)**
ICSD-244477										
RbCl (Rb[6], —)	0.78	0.58	6.8		*3.44*				108	
ICSD-22166										
Sr_3AlO_4F (Sr2[8], —)	0.79	0.75	6.8		*3.97*		1.07		102	
ICSD-253796										
$CsMgCl_3$(Mg[6],—)	0.79	0.91	6.8		4.37		1.41		115	
ICSD-54167										
$Ca_3B_2O_6$(Ca[8],—)	0.80	0.80			4.39		1.04		116	
ICSD-1894										
$La_2SrSc_2O_7$ (La/Sr2[12], Sc[6])	0.80	0.80		0.42	*3.64*		1.30		105	
ICSD-167047										
Sr_2YNbO_6 (Sr[8], Nb[6]) (↑)	0.80	0.85		0.55	*3.31*		0.35		117	
ICSD-35880					**3.63–3.77**		1.19			**1.86 (CT)**
KBr (K[6], —)	0.81	0.66	6.6		*3.35–3.40*				108 and 118	
ICSD-187220										
$Sr_2Y_8(SiO_4)_6O_2$ (Sr/Y1[9], Y1[9])	0.81	0.83		0.36	*3.32, 3.64*		0.32		119	
$Sr_2Y_8(SiO_4)_6O_2$ (Sr/Y1[9], Y2[7])				0.32	*3.59, 3.76*		1.08			
Ref. 119										
$CaGa_2O_4$ (Ca[8], Ga1[6]) (↑)	0.81	0.81		0.56	**3.68**		1.56		120	**2.17–2.34 (CT)**
$CaGa_2O_4$ (Ca[8], Ga2[6]) (↑)				0.53						
ICSD-155955										
$CaHfO_3$ (Ca[8], Hf[6]) (↑)	0.81	0.81	6.55	0.52	**4.03**		0.78		27	1.34 (A)
MP-754853										
$Ca_3Lu_2Ge_3O_{12}$ (Ca[8], Ge[4])	0.82	0.82		0.66	*3.26*		0.66		121	
Ref. 121										
$Ca_4ZrGe_3O_{12}$ (Ca[8], Ge[4])	0.82	0.82		0.66	*3.35*		0.50		122	
Ref. 123										
$CaZrO_3$ (Ca[8], Zr[6]) (↑)	0.82	0.82	6.55	0.49	**3.92–4.02**		0.74–0.83		124 and 125	1.11–1.20 (A)
ICSD-183758										
$CaSnO_3$ (Ca[8], Sn[6]) (↑)	0.82	0.82	6.7		**4.07**		0.63		124 and 125	1.23 (A)
ICSD-193658				0.53		**4.03–4.07**		1.33		**1.93 (CT)**
Ca_2MgWO_6 (Ca[8], W[6]) (↑)	0.82	0.82		0.54		**3.67**		1.42	126	**2.07 (CT)**
ICSD-155217										
$CaTiO_3$ (Ca[8], Ti[6]) (↑)	0.83	0.83	6.7	0.54		*3.35*		1.13	3, 125 and 127	
ICSD-165801						**3.65–3.87**		1.46–1.64		**2.10–2.28 (CT)**
$LiCa_3MgV_3O_{12}$ (Ca[8], V[4]) (↑)	0.83	0.83	7[a]	0.78		**3.64**		1.43	128	**2.19 (CT)**
ICSD-259239										
La_2MgGeO_6(La1[9],Ge[6])(↑)	0.84	0.84		0.55	4.13–4.24		1.03–1.23		129 and 130	**1.71–1.91 (CT)**
ICSD-97016										
SrY_2O_4 (Sr[8], Y2[6])	0.84	0.80		0.39	*3.76*		0.74		131	
ICSD-14741										
$Mg_3Y_2Ge_3O_{12}$ (Y/Mg1[8],Ge[4]) (≈)	0.84	0.91		0.67	**4.27**		0.42		132	0.36 (A)
ICSD-36636										
La_2MgGeO_6(La2[9],Ge[6]) (↑)	0.86	0.86		0.56	4.13–4.24		1.03–1.23		129 and 130	**1.68–1.88 (CT)**
ICSD-97016										
$Ba_3Lu_2B_6O_{15}$ (Ba[9], Lu1[6])	0.86	0.76		0.37	*3.40*		0.40		133	
$Ba_3Lu_2B_6O_{15}$ (Ba[9], Lu2[6])				0.38						
ICSD-27026										
KI (K[8], —)	0.89	0.74	6.25		3.26				108	
ICSD-9008654										
$Lu_3Al_5O_{12}$ (Lu[8],Lu[8]) (↓)	0.89	0.99	6.77		**4.56**–*4.63*		0.42–0.55		27, 38, 42,	0.11 (A)
$Lu_3Al_5O_{12}$ (Lu[8],Bi[8])(pairs)				0.37		**4.60**		2.00	134 and 135	**1.47 (CT)**

Table 5 (*continued*)

	Crystal lattice (doping site [CN], metal acceptor [CN′])									
					E_{exp} (Bi^{3+})		S			
Crystal database reference	he(M)	he(Bi^{3+})	U	$\frac{\chi_{CN'}}{d}$	A type	**CT type**	A type	**CT type**	Ref.	ΔE(Bi^{3+})
ICSD-17789				0.35		**4.58**		1.92		
$Y_3Ga_5O_{12}$ (Y[8],Ga[6]) (≈)	0.91	0.99	6.77		**4.35–4.36**		0.47–0.49		27, 35 and 136	0.39–0.40 (A)
$Y_3Ga_5O_{12}$ (Y[8],Bi[8])(pairs)				0.45		**4.07**		1.49		**1.41 (CT)**
ICSD-185862				0.34		**4.26–4.29**		1.50–1.72		
$SrZrO_3$ (Sr[8],Zr[6]) (≈)	0.91	0.87		0.47	**4.09**		1.22		137 and 138	1.41 (A)
ICSD-173398						**4.09**		1.22		
$Sr_3B_2O_6$(Sr[8],—)	0.92	0.86		—	4.57		1.17		116	
ICSD-93395										
La_3BWO_9 (La[9], W[6]) (↑)	0.92	0.92	6.9	0.60	**3.54**		1.40		139	**2.25 (CT)**
ICSD-39809										
$SrSnO_3$ (Sr[8],Sn[6]) (↑)	0.93	0.89		0.51	**4.13**		0.93		138	1.32 (A)
ICSD-262981						**4.13**		1.28		**1.67 (CT)**
$LaBO_3$(La[9],La[9]) (↓)	0.93	0.93	6.93	0.30	4.62	**4.53**	1.16	1.84	23 and 140	0.31 (A)
$LaBO_3$ (La[9],Bi[9])(pairs)				0.33						
ICSD-15383										
$YAlO_3$ (Y[8],Y[8]) (↓)	0.93	1.02	6.81	0.36	**4.43**		0.67		141	0.04 (A)
ICSD-191383										
$Y_3Al_5O_{12}$ (Y[8],Y[8]) (↓)	0.94	1.02	6.77		4.53–*4.64*		0.49–0.58		27–29, 38,	0.21 (A)
$Y_3Al_5O_{12}$ (Y[8],Bi[8])(pairs)				0.34		**4.55**		1.92	42 and 135	1.18 (A)
ICSD-280104				0.35		**4.32–4.52**		1.44–1.88		
Sr_2GdGaO_5(Sr1/Gd1[8],Ga[4]) (↓)	0.94	0.95		0.53	3.97		1.34		27	0.33 (A)
ICSD-1183										
$CaLaMgTaO_6$ (Ca/La[8], Ta[6]) (↑)	0.94	0.94	6.8[a]	0.58	**3.73**		0.93		142	**1.64 (CT)**
ICSD-160283										
$CaNb_2O_6$ (Ca[8], Nb[6]) (↑)	0.95	0.95	6.8	0.53		**3.81**		1.28	3 and 5	**1.70 (CT)**
ICSD-15208										
$Sr_3Lu_2Ge_3O_{12}$ (Sr[8], Ge[4])	0.95	0.90		0.66	*3.18*		0.52		143	
Ref. 143										
$Sr_3Y(BO_3)_3$ (Sr[8], Y1[6])	0.95	0.90		0.33	*3.35*		0.36		144	
$Sr_3Y(BO_3)_3$ (Sr[8], Y2[6])				0.38	*3.65*		1.22			
ICSD-246230										
Ba_2GdNbO_6 (Ba[8], Nb[6]) (↑)	0.95	0.85		0.52	*3.28*		0.61		145	
ICSD-172405					**3.93**					
Ba_2YGaO_5 (Ba1[8], Ga[4])	0.95	0.85		0.58		*3.54*		1.43	146	
Ref. 146										
$Gd_3Ga_5O_{12}$ (Gd[8],Ga[6]) (≈)	0.96	1.03	6.77	0.45		**4.00**		1.42	27, 35, 147	1.29 (A)
$Gd_3Ga_5O_{12}$ (Gd[8],Bi[8])(pairs)				0.34		**4.28**		1.74	and 148	**1.61 (CT)**
ICSD-184391						**4.10**		1.49		
Gd_2ZnTiO_6 (Gd[8], Ti[6])	0.97	1.04		0.50	*3.30*		0.34		149	
ICSD-251934										
$ScPO_4$ (Sc[8], Sc[8]) (↓)	0.97	1.15		0.36	**4.96**		2.14		150	1.36 (A)
Ref. 150										
Y_3TaO_7 (Y1[8]),Ta[6])(≈)	0.97	1.04		0.51	**3.87**		1.21		151	1.40 (A)
$LaVO_4$ (La[9], V[4]) (↑)	0.98	0.98	6.8, 7[a]	0.74		**3.82**		1.53	5	**1.83 (CT)**
ICSD-8294										
$GdAlO_3$(Gd[8],—)	0.98	1.05	6.75	—	4.24		0.52		31	
$GdAlO_3$ (Gd[8],Bi[8])(pairs)				0.35		**4.27**		1.69		
ICSD-59848										
$Lu_2Ge_2O_7$(Lu[8],Ge[6]) (↑)	0.98	1.08		0.52	4.35		0.89		152	1.09 (A)
ICSD-7221										
$SrLaGa_3O_7$ (Sr/La[8], Ga1[4])	0.98	0.95		0.44	*3.73*		1.11		153	
$SrLaGa_3O_7$ (Sr/La[8], Ga2[4])				0.54						
ICSD-109449										
$CaMoO_4$ (Ca[8],Mo[4])(≈)	0.99	0.99	7.05[a]	0.68	3.87		1.76		3, 5 and 154	**1.67 (CT)**
ICSD-193793					**4.32–4.38**	*3.83*		1.52		
$CsCdBr_3$[Cd[6],—)	0.99	1.02	6.6		4.33		2.21		115	
ICSD-87261										
$CaWO_4$ (Ca[8],W[4]) (≈)	1.00	1.00	7.1, 7.05[a]	0.69		**4.32–4.40**		1.67–1.65	155 and 156	**1.63–1.61 (CT)**
ICSD-33179										
$Y_2Ti_2O_7$ (Y[8],Ti[6]) (≈)	1.00	1.09	6.7	0.47		**3.86**		1.57	39	**1.48 (CT)**
ICSD-191622										
$LaOBr$(La[8],La[8]) (↓)	1.00	1.00	6.4			4.26		1.76	112	
MP-23023										
$CaSO_4$(Ca[8],—)	1.00	1.00			*3.50*		0.22		157	
ICSD-183917					4.09		0.81			
					5.10					
$LiCaBO_3$(Ca[7],—)	1.02	1.02	6.9		4.08		0.80		158	

Table 5 (*continued*)

	Crystal lattice (doping site [CN], metal acceptor [CN′])									
					E_{exp} (Bi^{3+})		S			
Crystal database reference	*he*(M)	*he*(Bi^{3+})	*U*	$\frac{\chi_{CN'}}{d}$	A type	**CT type**	A type	**CT type**	Ref.	ΔE(Bi^{3+})
ICSD-99386										
$BaLa_2ZnO_5$(La[8],La[8]) (↓)	1.02	1.02	6.55	0.35	3.87				109	
ICSD-88598										
$LaPO_4$ (La[9], La[9]) (↓)	1.03	1.03	7.18	0.31		**5.17**		2.41	159	**1.49 (CT)**
ICSD-431743										
$YNbTiO_6$ (Y[8], Ti[6]) (↑)	1.03	1.11	6.8	0.55		**3.64**		1.39	14	**1.72 (CT)**
$YNbTiO_6$ (Y[8], Nb[6]) (↑)				0.59						**1.95 (CT)**
ICSD-100175										
Y_2WO_6 (Y1[8], W[6]) (↑)	1.03	1.11	6.9	0.62		**3.59**		1.24	1	**1.97 (CT)**
ICSD-20955										
Gd_2GaSbO_7 [Gd[8],Ga[6]) (≈)	1.04	1.12	6.7	0.43	**4.27**		0.92		160	0.57 (A)
Gd_2GaSbO_7[Gd[8],Sb[6])				0.53						1.13 (A)
ICSD-201864										
$Sr_2Ga_2SiO_7$(Sr[8],Ga1[4])	1.05	1.00		0.45	*3.71*–3.92		1.07-1.25		161	0.08–0.26 (A)
$Sr_2Ga_2SiO_7$(Sr[8],Ga2[4])				0.54						
Ref. 161										
$LaYO_3$(La[8],Y[6])(↓)	1.05	1.05		0.39	3.76		1.25		162	0.75 (A)
ICSD-419666										
$LaScO_3$ (La[8], Sc[6])	1.05	1.05		0.42	*3.71*		0.82		137, 163 and 164	
ICSD-242159					*3.30*		0.35			
					3.60		1.10			
					4.54					
La_2ZnTiO_6 (La[8], Ti[6]) (≈)	1.05	1.05		0.53	**3.54**		0.50		165	
ICSD-174571										
Lu_2WO_6 (Lu2[8], W[6]) (↑)	1.05	1.13		0.62		**3.59**		1.16	166	**1.87 (CT)**
ICSD-161694										
La_2MgTiO_6 (La[8], Ti[6]) (≈)	1.05	1.05		0.52	**3.64**		0.76		167	1.00 (A)
ICSD-152284										
La_2LiSbO_6(La[8],La[8])[c] (↓)	1.05	1.05		0.32	3.96		0.92		168	0.03 (A)
ICSD-72202										
La_2LiNbO_6 (La[8],Nb[6]) (↑)	1.06	1.06		0.45	**3.76**		0.94		169	1.33 (A)
ICSD-230282										
$LaInO_3$ (La[8], In[6])	1.06	1.06	6.73	0.44	*3.65*		0.68		103	
ICSD-281549										
$LuVO_4$ (Lu[8],V[4]) (↑)	1.06	1.17	6.8, 7.0[a]	0.76		**3.73–3.76**		1.61–1.60	45 and 170	**1.82–1.83 (CT)**
ICSD-19163										
Gd_2GeO_5(Gd1[8],Ge[4]) (↓)	1.06	1.13	6.8	0.63	4.00		1.25		171	0.66 (A)
ICSD-61372										
$Ca_2B_2O_5$(Ca2[7],—)	1.07	1.07		-	4.23		0.76		116	
ICSD-190680										
Y_2WO_6 (Y2[8], W[6]) (↑)	1.07	1.14	6.9	0.62		**3.59**		1.24	1	**1.94 (CT)**
ICSD-20955										
$YNbO_4$ (Y[8],Nb[6]) (≈)	1.08	1.16	6.8	0.51		**4.00**		1.34	46, 172 and 173	**1.39 (CT)**
ICSD-239207						**4.06**		1.65		**1.70 (CT)**
						4.09		1.56		**1.61 (CT)**
$LuPO_4$ (Lu[8], Lu[8])	1.08	1.20	7.08	0.36	*5.51*		0.26		30	
$LuPO_4$ (Lu[8], Bi[8]) *(pairs)*				0.34		*5.51*		1.79		
ICSD-201133						*7.3*				
$CaYBO_4$ (Ca[7],Y[7]) (↓)	1.09	1.09	6.8	0.36	**4.44**				17	
ICSD-93388										
$Ba_2ZnGe_2O_7$ (Ba[8], Ge[4])	1.09	0.95		0.65	*3.42*		1.04		174	
$Ba_2ZnGe_2O_7$ (Ba[8], Zn[4])				0.35						
ICSD-8846										
$PbWO_4$ (Pb[8],W[4]) (≈)	1.10	1.03	7, 7.05[a]	0.68		**4.00**		1.80	43	**1.65 (CT)**
ICSD-195229										
YVO_4 (Y[8],V[4]) (≈)	1.10	1.19	6.8, 7.0[a]	0.76		**3.55**		1.37	1, 34, 41, 44,	**1.56 (CT)**
ICSD-78074						**3.59**		1.42	45, 175 and 176	**1.60 (CT)**
						3.64		1.42-1.46		**1.60–1.64 (CT)**
						3.67		1.54		**1.72 (CT)**
						3.72		1.54		**1.72 (CT)**
						3.78		1.59		**1.77 (CT)**
$YTaO_4$ (Y[8],Ta[6]) (↑)	1.11	1.20	6.8	0.54	4.27		1.32		5 and 177	**1.50 (CT)**
ICSD-238536						**3.61**		1.20		1.38 (A)
$MgGeO_3$(Mg1[6],Ge1[4]) (↓)	1.11	1.24	6.75	0.58	4.26		0.82		178	0.21 (A)
$MgGeO_3$(Mg1[6],Ge2[4]) (↓)				0.61						0.01 (A)
ICSD-35533										
$La_2Zr_2O_7$ (La[8],Zr[6]) (≈)	1.12	1.12	6.55	0.420.33	4.27		1.11		37	0.70 (A)

Table 5 (*continued*)

	Crystal lattice (doping site [CN], metal acceptor [CN′])									
					E_{exp} (Bi^{3+})		S			
Crystal database reference	he(M)	he(Bi^{3+})	U	$\chi_{CN'}/d$	A type	**CT type**	A type	**CT type**	Ref.	ΔE(Bi^{3+})
$La_2Zr_2O_7$ (La[8], La[8]) (↓)						**5.06**		**2.65**		**1.73 (CT)**
ICSD-253061										
YPO_4 (Y[8], Y[8])	1.12	1.22	7.09	0.33	*5.46–5.51*		0.34–0.43		30	
YPO_4 (Y[8], Bi[8]) (pairs)				0.34		*5.46–5.56*		1.75		
ICSD-79754						*7.1*				
$GdNbO_4$ (Gd[8],Nb[6]) (≈)	1.13	1.21	6.8	0.51		**3.84**		1.20	5 and 172	1.20 (A)
ICSD-20408						**4.07**		1.30		1.30 (A)
$Y_2Sn_2O_7$ (Y[8], Sn[6]) (≈)	1.13	1.17	6.75	0.46	**4.43**		0.70		26	0.46 (A)
$Y_2Sn_2O_7$ (Y[8],Bi[8])(pairs)				0.36		**4.32**		1.89		
ICSD-160115										
La_2GeO_5(La1[8],Ge[4]) (↓)	1.14	1.14		0.62	3.92		1.34		179	0.68 (A)
ICSD-59727										
La_4GeO_8(La1[8],Ge[4]) (↓)	1.14	1.14		0.54	*3.18*		1.11		180	
MP-1211987					3.81		1.23			0.05 (A)
$GdVO_4$ (Gd[8],V[4]) (≈)	1.15	1.23	6.8, 7.0[a]	0.76		**3.65**		1.31	5, 45, 84, 175	**1.54 (CT)**
ICSD-19161						**3.67**		1.43	and 181	**1.66 (CT)**
						3.78		1.44		**1.67 (CT)**
						3.79		1.55–1.63		**1.78–1.86 (CT)**
$MgGeO_3$ (Mg2[6], Ge1[4])	1.15	1.31	6.75	0.54	*4.26*		0.82		178	
$MgGeO_3$ (Mg2[6], Ge2[4])				0.67						
ICSD-35533										
Lu_2WO_6 (Lu1[8], W[6]) (↑)	1.16	1.19		0.62		**3.59**		1.16	166	**1.80 (CT)**
ICSD-161694										
Lu_2SiO_5(Lu2[7],Lu1[7]) (↓)	1.17	1.28	6.83	0.38	4.20		0.75		182 and 183	0.05 (A)
Lu_2SiO_5(Lu2[7],Lu2[7]) (↓)				0.40		4.05		1.75–1.85		1.04 (A)
Lu_2SiO_5(Lu2[7],Bi1[7])(pairs)				0.37		3.55		0.80		
ICSD-89624										
$ZnTiO_3$ (Zn[6], Ti[6])	1.17	1.32		0.55	*4.46*		1.01		184	
ICSD-262709						*3.28*		0.83		
$LiYSiO_4$ (Y[7], Y[7])	1.18	1.28	6.91	0.35	*4.43, 4.37*[b]				17 and 74	
ICSD-34079							0.86			
Ca_2MgWO_6 (Mg[6],W[6]) (≈)	1.19	1.35		0.54		**3.67**		1.42	126	**1.44 (CT)**
ICSD-155217										
Y_2SiO_5 (Y1[7], Y2[7]) (↓)	1.19	1.27	6.8	0.38	**4.50**		0.94		185	0.14 (A)
ICSD-291358										
$ScVO_4$ (Sc[8],V[4]) (≈)	1.20	1.40[d]	6.8, 7.0[a]		**3.44**		0.69		175, 186 and 187	0.99 (A)
ICSD-257652				0.81		**3.44–3.46**		1.50–1.51		**1.81 (CT)**
La_3SbO_7 (La1[8], Sb[6])	1.20	1.20		0.51	*3.54*		1.16		188	
ICSD-188501										
Cs_2HfCl_6(Hf[6],—)	1.21	1.40			3.54		0.85		94	
MP-1120741										
$Mg_3Y_2Ge_3O_{12}$ (Mg2[6], Ge[4])	1.21	1.38		0.59	*4.27*		0.42		132	
ICSD-36636										
Cs_2SnCl_6(Sn[6],—)	1.22	1.43		-	3.55		0.83		189	
MP-608555										
$CaLaMgTaO_6$ (Mg[6],Ta[6]) (≈)	1.22	1.38	6.8[a]	0.58	**3.73**		0.93		142	1.15 (A)
ICSD-160283										
$ZnGa_2O_4$(Ga[6],Ga[6]) (↓)	1.23	1.55	6.8	0.43	3.44		0.42		190	0.37 (A)
$ZnGa_2O_4$ (Ga[6], Ga[6]) (↓)						**4.43**		1.85		1.06 (A)
ICSD-237886										
La_2ZnTiO_6(Zn[6],Ti[6]) (↓)	1.23	1.33		0.42	3.54		0.50		165	0.13 (A)
ICSD-174571										
Cs_2NaYCl_6(Y[6],—)	1.24	1.32	6.67	—	3.84		0.12		191	
ICSD-245353										
La_2MgTiO_6(Mg[6],Ti[6]) (↓)	1.24	1.40		0.42	3.64		0.76		167	0.06 (A)
ICSD-152284										
La_2MgGeO_6 (Mg[6], Ge[6]) (↓)	1.25	1.42		0.46	**4.13–4.24**		1.03–1.23		129 and 130	0.53–0.73 (A)
ICSD-97016										
Gd_2GeO_5(Gd2[7],Ge[4]) (↓)	1.28	1.36	6.8	0.65	4.00		1.25		171	0.67 (A)
ICSD-61372										
$LiLaP_4O_{12}$ (La[8], La[8]) (↓)	1.28	1.28	7.29	0.23		**5.27–5.40**		2.25–2.51	40 and 192	0.63–0.89 (A)
$LiLaP_4O_{12}$ (La[8], Bi[8]) (pairs)						*4.95*		2.78		
ICSD-421741										
$Ca_5(VO_4)_3F$ (Ca2[7], V[4]) (≈)	1.28	1.28	7[a]	0.78		**3.40**		1.63	33	**1.89 (CT)**
ICSD-172997										
LaP_3O_9 (La[8], La[8]) (↓)	1.28	1.28	7.26	0.30	**5.27**		2.55		91	1.33 (A)
ICSD-202640										

Table 5 (*continued*)

	Crystal lattice (doping site [CN], metal acceptor [CN′])									
					E_{exp} (Bi^{3+})		S			
Crystal database reference	he(M)	he(Bi^{3+})	U	$\frac{\chi_{CN'}}{d}$	A type	**CT type**	A type	**CT type**	Ref.	ΔE(Bi^{3+})
$CaYBO_4$ (Y[7], Y[7]) (↓)	1.29	1.39	6.8	0.37	**4.44**				17	
ICSD-93388										
$LaNbO_4$ (La[8],Nb[6]) (≈)	1.29	1.29	6.8	0.51		**3.93–4.07**		1.40	5 and 172	1.31 (A)
$BaScO_2F$(Sc[6],Sc[6]) (↓)	1.30	1.39		0.33	*2.99*				98	
ICSD-19622					3.44		0.54			0.21 (A)
					4.22					
$CaGa_2O_4$(Ga1-2[6],Ga1-2[6]) (↓)	1.30	1.54		0.49	3.68		1.56		120	1.12 (A)
ICSD-155955										
Cs_2ZrCl_6(Zr[6],—)	1.30	1.50			3.49		0.77		96	
ICSD-26695										
Y_2SiO_5 (Y2[7], Y2[7]) (↓)	1.30	1.39	6.8	0.38	**4.50**		0.94		185	0.02 (A)
ICSD-291358										
$Sr_2Y_8(SiO_4)_6O_2$(Y2[7],Y1[9]) (↓)	1.31	1.39		0.31	*3.32,* 3.64		0.32		119	< 0.94 (A)
$Sr_2Y_8(SiO_4)_6O_2$(Y2[7],Y2[7]) (↓)				0.32	3.59,3.76		1.08			
Ref. 119										
Lu_2SiO_5(Lu1[7],Lu1[7])	1.31	1.43	6.83	0.38	*4.20*		0.75		182 and 183	
Lu_2SiO_5(Lu1[7],Lu2[7])				0.38		4.05		1.75–1.85		0.79–0.89 (A)
Lu_2SiO_5(Lu1[7],Bi1[7])(pairs)				0.36		3.55		0.80		
Lu_2SiO_5(Lu1[7],Bi2[7])(pairs)				0.37						
ICSD-89624										
La_4GeO_8(La2[8],Ge[4]) (↓)	1.31	1.31		0.64	*3.18*		1.11		180	
MP-1211987					3.81		1.23			0.51 (A)
La_2SO_6 (La[8], La[8])	1.32	1.32		0.34	*4.45*		1.36		17	
ICSD-66823										
$NaScO_2$(Sc[6],Sc[6]) (↓)	1.33	1.56	6.4	0.43	3.43		0.18		193	0.61 (A)
ICSD-25729										
Gd_2ZnTiO_6 (Zn[6], Ti[6])	1.33	1.50		0.43	*3.30*		0.34		149	
ICSD-251934										
Y_3TaO_7 (Y2[7]),Ta[6])(≈)	1.34	1.45		0.51	**3.87**		1.43		151	1.18 (A)
CaO(Ca[6],—)	1.35	1.35	6.31		3.47–3.50		0.34–0.40		194–197	
ICSD-7200686										
Lu_2WO_6 (Lu3[7],W[6]) (↑)	1.35	1.46		0.62		**3.59**		1.16	166	**1.53 (CT)**
ICSD-161694										
$GaBO_3$ (Ga[6], Ga[6]) (↓)	1.36	1.69	6.9		**4.49**		0.21		198	0.75 (A)
ICSD-429062				0.42		**4.44**		*1.51*		0.54 (A)
Y_2WO_6 (Y3[7],W[6]) (↑)	1.37	1.47	6.9	0.62		**3.59**		1.24	1	**1.60 (CT)**
ICSD-20955										
$Cs_2NaBiCl_6$	1.37	1.37			3.69[b]		1.84		199	
ICSD-59195										
$Li_4SrCa(SiO_4)_2$(Ca[6],—)	1.37	1.37	6.8		4.00		0.74		113	
ICSD-259898										
$Ca_4GdO(BO_3)_3$(Ca2[6],—)	1.37	1.37	6.8		4.00		0.82		200	
ICSD-132431										
Ba_2YGaO_5(Ba2[7],Ga[4]) (↓)	1.37	1.58		0.51	3.54			1.43	146	0.31(A)
Ref. 146										
$LiScO_2$(Sc[6],Sc[6]) (↓)	1.37	1.62	6.4	0.43	3.92		0.86		193	0.01 (A)
ICSD-31316										
$LiScGeO_4$ (Sc[6], Ge[4]) (↓)	1.39	1.61		0.69	**4.13**		0.73		201	0.05 (A)
ICSD-62481					**4.43**					
$CaSb_2O_6$(Ca[6],Sb[6]) (≈)	1.39	1.39		0.50	**3.64–3.69**[b]		0.85–1.00		2, 12 and 202	0.60–0.76 (A)
ICSD-74539										
$Ca_2B_2O_5$ (Ca1[6], —)	1.39	1.39			*4.23*		0.76		116	
ICSD-190680										
$Ca_4YO(BO_3)_3$(Ca2[6],Y[6]) (↓)	1.40	1.40	6.82	0.34	4.00		0.74		203	0.41 (A)
ICSD-132433										
$LaGaO_3$ (La[7],Ga[6]) (↓)	1.40	1.40	6.7	0.48	**4.02–4.03**		0.70–0.74		204 and 205	0.33–0.37 (A)
ICSD-259867										
La_2GeO_5(La2[7],Ge[4]) (↓)	1.40	1.40		0.64	3.92		1.34		179	0.54 (A)
ICSD-59727										
$Sr_3Y(BO_3)_3$(Y1[6],Y2[6]) (↓)	1.40	1.47		0.28	3.35		0.36		144	1.20 (A)
ICSD-246230					3.65		1.22			0.34 (A)
Sc_2O_3(Sc1-2[6],Sc1-2[6]) (↓)	1.41	1.64	6.45	0.42	3.33		0.28		175 and 206	0.64 (A)
Sc_2O_3 (Sc1-2[6], Sc1-2 [6]) (↓)					3.67–3.70	**4.60**	0.65–1.10	*2.14*		0.18–0.27 (A)
ICSD-169172										1.22 (A)
La_2O_3(La[7],La[7]) (↓)	1.42	1.42	6.45		4.03–4.07		1.40–1.49		25, 175 and 207	0.28–0.37 (A)
La_2O_3 (La[7],Bi[7])(pairs)				0.35		**4.98**		*2.40*		1.28 (A)
ICSD-56166				0.37		**3.65**		1.47		

Table 5 (*continued*)

	Crystal lattice (doping site [CN], metal acceptor [CN′])									
					E_{exp} (Bi^{3+})		S			
Crystal database reference	he(M)	he(Bi^{3+})	U	$\frac{\chi_{CN'}}{d}$	A type	**CT type**	A type	**CT type**	Ref.	ΔE(Bi^{3+})
$CaYGaO_4$(Ca[6],Ga[4]) (↓)	1.42	1.42		0.57	3.35		0.53		208	0.73 (A)
Ref. 208					3.72					
Cs_2NaYBr_6(Y[6],—)	1.42	1.50	6.53		3.36		0.28		209	
ICSD-65733										
$Ca_4GdO(BO_3)_3$(Ca1[6],—)	1.43	1.43	6.8		4.00		0.82		200	
ICSD-132431										
$Ca_4YO(BO_3)_3$(Ca1[6],Y[6]) (↓)	1.44	1.44	6.82	0.37	4.00		0.74		203	0.28 (A)
ICSD-132433										
$Cs_2NaLaCl_6$(La[6],—)	1.45	1.45	6.67		3.90		0.27		209	
ICSD-425945										
$SrGa_2O_4$(Sr2[6],Ga1[4])(↓)	1.45	1.40		0.53	3.76		1.47		210	0.04 (A)
$SrGa_2O_4$(Sr2[6],Ga2[4])(↓)				0.46						0.49 (A)
$SrGa_2O_4$(Sr2[6],Ga3[4])(↓)				0.47						0.43 (A)
$SrGa_2O_4$(Sr2[6],Ga4[4])(↓)				0.54						0.02 (A)
ICSD-91221										
La_3SbO_7 (La2[7],Sb[6]) (≈)	1.48	1.48		0.56	**3.54**		1.16		188	1.17 (A)
ICSD-188501										
SrO(Sr[6],—)	1.48	1.40	6.4		3.38		0.53		194 and 211	
ICSD-9008727										
$Ca(OH)_2$(Ca[6],—)	1.49	1.49			3.73 (sh)		0.89		212	
ICSD-191851					*4.16*					
$ScBO_3$ (Sc[6], Sc[6])	1.50	1.75	6.86	0.36	*4.38*		0.12		23	
ICSD-194864										
$LaScO_3$(Sc[6],La[8]) (↓)	1.51	1.73		0.38	3.71		0.82		137, 163 and 164	0.40–0.63 (A)
LaScO3 (Sc[6]), Sc[6]) (↓)					3.30		0.35			0.84–1.07 (A)
ICSD-242159					3.60		1.10			0.12–0.35 (A)
					4.54					
$BaGd_2O_4$(Ba[8],—)	1.51	1.36			3.68		0.81		213	
ICSD-427487										
$BaSc_2O_4$ (Sc2[6], Sc1[6])	1.51	1.70		0.45	*3.91*		1.81		101	
$BaSc_2O_4$ (Sc2[6], Sc2[6])				0.43	**4.63**					
$BaSc_2O_4$ (Sc2[6], Sc3[6])				0.41						
ICSD-35592										
$SrGa_2O_4$(Sr1[6],Ga1[4]) (↓)	1.52	1.47		0.58	3.76		1.47		210	0.21 (A)
$SrGa_2O_4$(Sr1[6],Ga2[4]) (↓)				0.61						0.40 (A)
$SrGa_2O_4$(Sr1[6],Ga3[4]) (↓)				0.63						0.53 (A)
$SrGa_2O_4$(Sr1[6],Ga4[4]) (↓)				0.47						0.49 (A)
ICSD-91221										
$NaLuO_2$(Lu[6],Lu[6]) (↓)	1.52	1.67	6.5	0.42	3.52		0.28		193	0.67 (A)
MP-755311										
$LaInO_3$(In[6],La[8]) (↓)	1.53	1.71	6.73	0.44	3.65		0.68		103	0.19 (A)
$LaInO_3$(In[6],In[6]) (↓)				0.35						
ICSD-281549										
$LiLuGeO_4$ (Lu[6], Ge[4]) (↓)	1.54	1.67		0.68	**4.06**[b]		0.52		214	0.27 (A)
Ref. 214										
$SrSb_2O_6$(Sr[6],Sb[6]) (↓)	1.55	1.50		0.49	3.47		1.17		2	0.77 (A)
ICSD-74540					3.66		0.57			0.17 (A)
$NaLuGeO_4$ (Lu[6], Ge[4]) (↓)	1.55	1.69	6.9	0.67	**4.10**[b]		1.05		74	0.18 (A)
ICSD-54196										
$CaYGaO_4$(Y[6],Ga[4]) (↓)	1.55	1.66		0.55	3.35		0.53		208	1.09 (A)
Ref. 208					3.72					
$Sr_3Lu_2Ge_3O_{12}$(Lu[6],Ge[4]) (↓)	1.55	1.67		0.57	3.18		0.52		143	0.98 (A)
Ref. 143										
$NaYO_2$(Y[6],Y(6) (↓)	1.56	1.68	6.45	0.39	3.51		0.29		193	0.85 (A)
ICSD-291907										
$LiLuO_2$(Lu[6],Lu[6]) (↓)	1.56	1.71	6.5	0.42	3.76		1.34		193	0.36 (A)
MP-754605										
$BaSc_2O_4$ (Sc1[6], Sc1[6])	1.57	1.78		0.34	*3.91*		1.09		101	
$BaSc_2O_4$ (Sc1[6], Sc2[6])				0.45	**4.63**					
$BaSc_2O_4$ (Sc1[6], Sc3[6])				0.32						
ICSD-35592										
Lu_2O_3(Lu2[6]=C2,Lu1-2[6]) (↓)	1.57	1.73	6.5	0.40	3.33		0.25		175	0.86 (A)
Lu_2O_3(Lu2[6]=C2,Lu1-2[6]) (↓)					3.61		1.19			0.08 (A)
ICSD-194467										
$InBO_3$(In[6], In[6]) (↓)	1.58	1.79	6.86	0.38	**4.37**		0.29		36 and 198	0.98 (A)
ICSD-75254						**4.37–4.43**		*1.27–1.33*		0.002–0.06 (A)
Lu_2GeO_5 (Lu1[6], Ge[4]) (↓)	1.58	1.71		0.64	**4.35**		0.39		215	0.69 (A)

Table 5 (*continued*)

Crystal database reference	Crystal lattice (doping site [CN], metal acceptor [CN′]) *he*(M)	*he*(Bi^{3+})	*U*	$\chi_{CN'}$/*d*	E_{exp} (Bi^{3+}) A type	E_{exp} (Bi^{3+}) CT type	*S* A type	*S* CT type	Ref.	ΔE(Bi^{3+})
MP-768388										
Lu_2O_3(Lu1[6]=S6,Lu1-2[6]) (↓)	1.59	1.72	6.5	0.40	3.33		0.25		175	0.85 (A)
ICSD-194467					3.61		1.19			0.09 (A)
MgS(Mg[6],—)	1.59	1.85	6.22		2.98		0.17		216	
ICSD-9008672										
$BaSc_2O_4$ (Sc3[6], Sc1[6])	1.59	1.79		0.32	*3.91*		1.09		101	
$BaSc_2O_4$ (Sc3[6], Sc2[6])				0.41	**4.63**					
$BaSc_2O_4$ (Sc3[6], Sc3[6])				0.42						
ICSD-35592										
Ba_2YGaO_5(Y[6],Ga[4]) (↓)	1.59	1.69		0.44	3.54			1.43	146	0.92 (A)
Ref. 146										
$NaYGeO_4$ (Y[6], Ge[4]) (↓)	1.60	1.71	6.9	0.66	**4.09**[b]		1.00		74	0.05 (A)
ICSD-85497										
$LiYO_2$(Y[6],Y[6]) (↓)	1.60	1.73	6.45	0.38	3.82		1.59		193	0.37 (A)
ICSD-45544										
Y_2O_3(Y1[6]=S6,Y1-2[6]) (↓)	1.60	1.72	6.46	0.37	3.31–3.32		0.27–0.31		175 and 206	1.00–0.99 (A)
Y_2O_3 (Y1[6] = S6, Y1–2 [6]) (↓)					3.52–3.73	**4.71**	0.70–1.20	*2.13*		0.07–0.57 (A)
ICSD-192862										0.85 (A)
$Ca_3Lu_2Ge_3O_{12}$(Lu[6],Ge[4]) (↓)	1.60	1.72		0.58	3.26		0.66		121	0.82 (A)
Ref. 121										
Y_2O_3(Y2[6]=C2,Y1-2[6]) (↓)	1.61	1.75	6.46	0.37	3.31–3.32		0.27–0.31		175 and 206	1.02–0.98 (A)
Y_2O_3 (Y2[6] = C2, Y1–2 [6]) (↓)					3.52–3.73	**4.71**	0.70–1.20	*2.13*		0.10–0.60 (A)
ICSD - 192862										0.83 (A)
$La_2SrSc_2O_7$(Sc[6],Sc[6]) (↓)	1.61	1.85		0.33	3.64		1.30		105	0.09 (A)
ICSD-167047										
$CaBaZn_2Ga_2O_7$(Ca[6],Ga1[4]) (↓)	1.61	1.61		0.47	3.81		1.64		99	0.45 (A)
$CaBaZn_2Ga_2O_7$(Ca[6],Ga2[4]) (↓)				0.48						0.39 (A)
ICSD-194164										
$Ca_4YO(BO_3)_3$ (Y[6], Y[6])	1.62	1.72	6.82	0.37	*4.00*		0.74		203	
ICSD-132433										
Lu_2GeO_5 (Lu2[6], Ge[4]) (↓)	1.62	1.75		0.62	**4.35**		0.39		215	0.85 (A)
MP-768388										
La_4GeO_8(La3[7],Ge[4]) (↓)	1.62	1.62		0.56	3.18		1.11		180	0.41 (A)
MP-1211987					3.81		1.23			0.29 (A)
Y_2GeO_5(Y2[6],Ge[4]) (↓)	1.62	1.72		0.58	3.81(sh)		0.45[e]		217	1.03 (A)
ICSD-260425					*4.13*					
					4.77					
La_2O_2S(La[7],—) (↓)	1.63	1.63			3.60		0.88		218	
ICSD-260145										
$Ca_4ZrGe_3O_{12}$(Ca/Zr2[6],Ge[4]) (↓)	1.64	1.76		0.58	3.35		0.50		122	1.01 (A)
Ref. 123										
$LiGdO_2$(Gd[6],—)	1.65	1.76	6.5		3.74		1.05		193	
ICSD-422651										
$NaGdO_2$(Gd[6],—)	1.65	1.75	6.5		3.61		0.39		193	
ICSD-97542										
$Sr_3Y(BO_3)_3$(Y2[6],Y1[6]) (↓)	1.65	1.76		0.28	3.35		0.36		144	1.20 (A)
ICSD-246230					3.65		1.22			0.60 (A)
SrY_2O_4(Y1[6],Y1-2[6]) (↓)	1.66	1.77		0.37	3.76		0.74		131	0.46 (A)
ICSD-14741										
$Ba_3Lu_2B_6O_{15}$(Lu1-2[6],Lu2-1)	1.66	1.78		0.23	3.40		0.40		133	
ICSD-27026										
$LuBO_3$(Lu[6], Lu[6])	1.66	1.81	6.94	0.36	*4.35*		0.37		23	
ICSD-194864										
$YAl_3B_4O_{12}$ (Y[6], Y[6])	1.66	1.77	7.04	0.22	*4.58–4.77*		0.36–0.49		219	
ICSD-187082										
Gd_2O_3(Gd2[6]=C2,—)	1.67	1.78	6.52		3.27–3.29		0.29–0.32		175, 220 and 221	
ICSD-230768					3.57–3.76		1.38–1.40			
CaZnOS (Ca[6], —)	1.67	1.78			3.39		0.83		222	
ICSD-245309										
$NaGdGeO_4$ (Gd[6], Ge[4]) (↓)	1.67	1.77		0.66	**4.10**[b]		0.85		223	0.15 (A)
ICSD-85496										
Y_2GeO_5 (Y1[6], Ge[4])	1.69	1.81		0.56	*3.81 (sh)*		0.45[e]		217	
ICSD-260425					*4.13*					
					4.77					
$LaYO_3$ (Y[6], La[8])	1.70	1.83		0.37	*3.76*		1.25		162	0.04 (A)
$LaYO_3$ (Y[6], Y[6])				0.32						0.33 (A)
ICSD-419666										

Table 5 (*continued*)

	Crystal lattice (doping site [CN], metal acceptor [CN′])									
					E_{exp} (Bi^{3+})		S			
Crystal database reference	*he*(M)	*he*(Bi^{3+})	U	$\frac{\chi_{CN'}}{d}$	A type	**CT type**	A type	**CT type**	Ref.	ΔE(Bi^{3+})
Gd_2O_3(Gd1[6]=S6,—) ICSD-230768	1.71	1.80	6.52		3.27–3.29 3.57–3.76		0.29–0.32 1.38–1.40		175, 220 and 221	
$Ca_4GdO(BO_3)_3$ (Gd[6], —) ICSD-132431	1.74	1.83	6.8		*4.00*		0.82		200	
$BaGd_2O_4$(Gd1[6],—) ICSD-427487	1.75	1.84			3.68		0.81		213	
$BaGd_2O_4$(Gd2[6],—) ICSD-427487	1.79	1.88			3.68		0.81		213	
$NaLaO_2$(La[6],La[6]) (↓) MP-755586	1.79	1.79	6.45	0.35	3.53		1.30		193	0.14 (A)
Sr_2YNbO_6(Y[6],Nb[6]) (↓) ICSD-35880	1.80	1.92		0.44	3.31 3.63–*3.77*		0.35 1.19		117	0.65 (A) 0.15 (A)
Ba_2GdNbO_6(Gd1[6],Nb[6]) (↓) ICSD-172405	1.87	1.96		0.43	3.28 *3.93*		0.61		145	0.31 (A)
CaS(Ca[6],—) ICSD-603165	1.96	1.96	6.17		3.01–3.02 3.57 *3.97*		0.27		194 and 195	
SrS(Sr[6],—) ICSD-659132	2.09	1.92	6.25		2.88		0.28		194 and 224	
CaSe(Ca[6],—) ICSD-41957	2.10	2.10	6.1, 6.22		*2.74* 3.22 *3.60*		0.24		195	
BaS(Ba[6],—) ICSD-52690	2.33	2.08	6.25		2.77		0.49		225	

sh: shoulder. [a] Value from ref. 226. [b] Average position of the lowest excitation doublet. [c] Bi–Sb MMCT not considered for this compound.[12] [d] Calculated based on a 7-fold coordinated site.[150] [e] Taken from shoulder.

in pairs), when present. A dash is mentioned otherwise or in case of non-purely oxidic compounds. When two kinds of metal acceptors are present in the crystal structure (*e.g.* $Y_3Ga_5O_{12}$, $Y_2Sn_2O_7$, Gd_2GaSbO_7, *etc.*), the CT transition of lowest energy is, in general, considered. Note that several compounds reported in the archival literature do not appear in Table 5 because their crystal structure is either not available (*e.g.* Ln_2SO_6 (Ln = Y, Lu),[17] $Sr_3YAl_2O_{7.5}$,[71] $BaLa_2Si_3O_{10}$,[72] Sr_2GdAlO_5,[27] $SrLaBO_4$,[73] $LiLuSiO_4$,[74] $Sr_3Lu(BO_3)_3$,[75] $LiYGeO_4$,[76] $Ba_3Sc_4O_9$,[77] $Ba_2Ga_2GeO_7$,[78] Gd_2MgTiO_6[79]) or too complex for the calculator that we have designed in this work (*e.g.* $Sr_2V_2O_7$ (ICSD-10330),[80] $Ba_3Y_4O_9$ (ICSD-230718),[81] K_2MgGeO_4 (MP-1212053),[82] $Y_4Al_2O_9$ (ICSD-63650)[83] $Ba_2Y_5B_5O_{17}$ (ICSD-8356),[84] $Na_3YSi_3O_9$ (ICSD-20774),[85] $Ca_3Al_4ZnO_{10}$ (ICSD-120597),[86] $CdSiO_3$ (ICSD-170703),[87] $Sr_3Sc_4O_9$,[88] AMB_5O_{10} with M = La, Gd, Y and A = Mg, Zn, and Cd,[89] $Li_2BaP_2O_7$ (ICSD-280927),[90] and MP_3O_9 with M = Sc, Y, Lu).[91] Compound Zn_2GeO_4:Bi[92] contains only tetrahedral sites that are not suitable for Bi^{3+} and was, for this reason, discarded. We also discarded YBO_3 that is confusingly described using different space groups. At last, Bi^{3+} doping in monovalent cation sites was not considered as a relevant option when divalent or trivalent cations are also present in the crystal structure. Table 5 also contains values of U and $\frac{\chi_{CN'}}{d}$, when available. According to P. Dorenbos,[93] the U value of a compound is a good indicator of the nephelauxetic effect, and we indeed note in Fig. 4 that U and *he* correlate, although with substantial scatter. One reason for such a scattering is that the U value of a compound, contrarily to *he*, does not consider the nature of the crystal site available for the dopant. More precisely, since the U value derives from the lanthanides spectroscopy, it is implicitly assumed that Bi^{3+} occupies the same crystal site as lanthanides in the considered host lattice. This, however, is not always an obvious option. It comes that a compound offering several possible doping sites for Bi^{3+} may have one value of U but several values of *he*. Another reason for the U–*he* discrepancy is that in closed-shell transition metal oxides like YVO_4 or $CaTiO_3$ and many others (see Table 5) the Bi^{3+} spectroscopy is primarily governed by charge transfer processes rather than by the nephelauxetic effect. The data reproduced with full squares in Fig. 4 pertain to compounds that do not contain d^0 or d^{10} cations. We see that the U-*he* correlation is now more consistent, although still imperfect.

For all compounds listed in Table 5, the experimental position, E_{exp}(Bi^{3+}), of Bi^{3+}-related bands (including Bi^{3+} pairs) and related Stokes shifts, S, are given as picked up from the archival literature. These bands are accordingly classified as "A type" or "CT type" transitions, where CT stands for MMCT or IVCT. Oxidic compounds, and their corresponding optical data, that show absorption in the CT state appear in bold in the Table. We have plotted in Fig. 5 the variation of E_{exp}(Bi^{3+}) against the E_{CT}(Bi^{3+}) values calculated by means of eqn (4) for compounds referenced to as CT-absorbers (graph 5(a)) and against *he*(Bi^{3+}) for compounds referenced to as A-absorbers (graph 5(b)). The plots evidence a convergence within ±0.35 eV in the first case (except for YPO_4 and $LuPO_4$) and ±0.35 eV in the second case using K = 0.555 and α = 0.24 in eqn (18). This set of (K, α) coefficients differs somehow from the values reported earlier.[53,54] The reason is that the values reported in

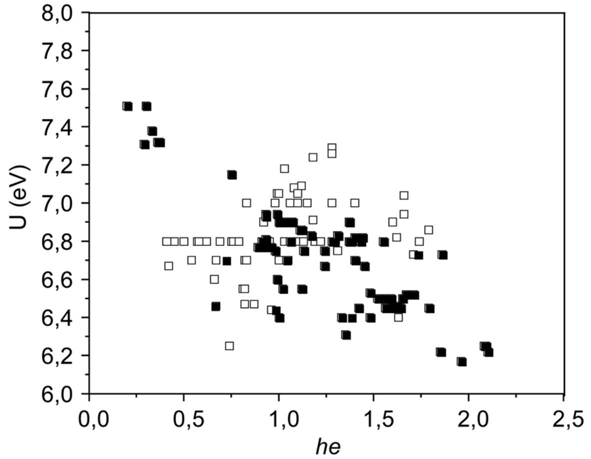


Fig. 4 The *U*–*he* correlation for Bi^{3+}-doped compounds. Full squares pertain to compounds that do not contain closed-shell transition metal cations in their crystal structure, see text.

these papers were obtained on chlorides and bromides only and were based on a much smaller sampling. We regard this new set of *K*, α values as the most relevant to date. Note that the ±0.35 eV accuracy that we retain for these models amounts twice the uncertainty that affects the spectroscopic data as they are picked up from the archival literature, see *e.g.* the case of YVO_4:Bi^{3+} in Table 5.

The compounds appear underlined in Table 5 when their corresponding excitation data (also underlined) obey eqn (18) ±0.35 eV. Since the *he*(M) and *he*(Bi^{3+}) values are sensitive to the structural attributes of the crystal sites, the spectral attributions made in Table 5 are site selective, *i.e.* we identify the nature of the crystal sites that accounts for the observed optical features.

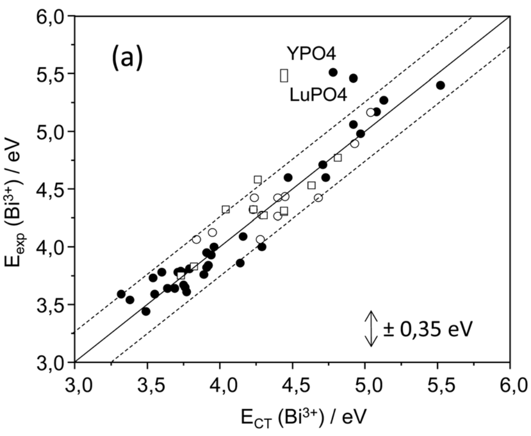
For compounds containing d^0 or d^{10} metal ions, the A and CT states co-exist. This is illustrated in Fig. 3. These compounds are indicated bold and underlined in Table 5 when their excitation data are reproduced by eqn (4) and (18) within ±0.35 eV. In situations where the A and CT states are separated by less than ±0.35 eV, then the A and CT states are considered as resonant, and the symbol (≈) is indicated after the compound's name. Otherwise, a vertical arrow pointing up (↑) is mentioned after the compound's name when the A state is calculated by more than 0.35 eV above the CT state, and reversely the arrow points down (↓) when the A state is calculated by more than 0.35 eV below the CT. When relevant, we give the $\Delta E(Bi^{3+})$ values calculated by means of eqn (5) at the last column of Table 5. At last, we indicate in italic the crystal sites that are *not reproduced* by the models. In this frame, the literature assignments are confirmed by the models when the data listed in column "A type" are underlined and those listed in column "CT type" (including Bi^{3+} pairs) are in bold. This holds for ≈80% of the compounds listed in the Table. We find that a $\Delta E(Bi^{3+})$ value of 1.40 ± 0.01 eV constitutes a threshold below which the compounds emit from the A state, and above which they emit from the CT state. The origin of emission for compounds located in the $\Delta E(Bi^{3+})$ range 1.39–1.41 eV must be examined on a case-by-case basis. It is noted that several A-emitting compounds listed in Table 5 possess a Stokes shift above 1 eV.

5. Discussion

Dealing with the last 20% of the literature assignments that are not reproduced, about 60% (≈12% of Table 5) are modeled based on an assignment that differs from the literature and the remaining amount (<9% of Table 5) is not reproduced by the models.

5.1. The unreproduced compounds

The unreproduced compounds are alkali halides, $SrLaGa_3O_7$, Sr_3AlO_4F, $Ba_2ZnGe_2O_7$, $YAl_3B_4O_{12}$, La_2SO_6, Gd_2ZnTiO_6, $ZnTiO_3$, $LuPO_4$, YPO_4, $ScBO_3$ and $LuBO_3$ in their calcite form. Concerning Gd_2ZnTiO_6 and $ZnTiO_3$, we suspect that the signals at 3.30 and 3.28 eV pertain to titanate units instead of Bi^{3+}. For La_2SO_6(La[8], La[8]):Bi, the 4.51 eV signal[17] possibly results from La → La MMCT; the difference between calculation and experiment is 0.39 eV *i.e.* 0.04 eV above our 0.35 eV tolerance.



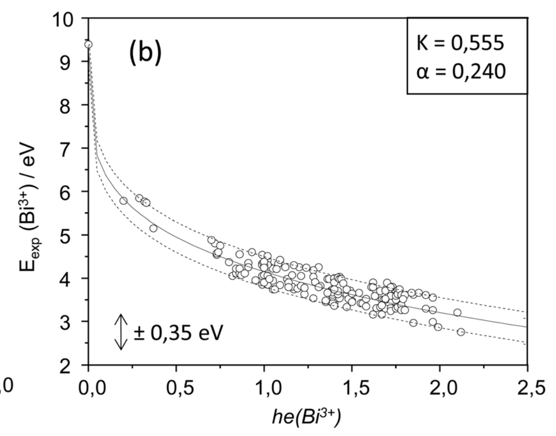


Fig. 5 $E_{exp}(Bi^{3+})$ plotted against $E_{CT}(Bi^{3+})$ for CT-absorbers (● = Bi^{3+}-to-d^0 MMCT, ○ = Bi^{3+}-to-d^{10} MMCT, □ = Bi^{3+}-to-Bi^{3+} IVCT) (a) and against *he*(Bi^{3+}) for A-absorbers (b).

5.1.1. The zircon phosphates YPO_4 and $LuPO_4$. The data related to YPO_4 and $LuPO_4$, and those of the other phosphates $ScPO_4$:Bi, $LaPO_4$:Bi, LaP_3O_9:Bi and $LiLaP_4O_{12}$:Bi that absorb in a similar spectral region, are summarized in the Table 6. The position of the C band ($^1S_0 \rightarrow {}^1P_1$ transition) appearing in this Table as $E_C(Bi^{3+})$ was estimated from the empirical equation:[21]

$$E_c(Bi^{3+}) = 3.236 + 2.29(E_A(Bi^{3+}) - 2.972)^{0.856} \quad (19)$$

The energy level structure corresponding to these compounds is depicted in Fig. 6. For all, A is the lower-lying state. This state is not observed on spectra. This was already pointed out by Boulon *et al.* in the case of $LaPO_4$:Bi.[159]

We find that the excitation of Bi^{3+} in YPO_4 and $LuPO_4$ is well reproduced by a C transition. The same holds for LaP_3O_9 and $LiLaP_4O_{12}$, although in these two cases, a La → La MMCT transition also matches. In the case of $ScPO_4$ and $LaPO_4$, only the Sc → Sc or La → La MMCT states seem to be involved in the excitation process. For $LaPO_4$, the $\Delta E(Bi^{3+})$ value of 1.49 eV suggests a CT character for the emission, *i.e.* the absorbing state is the emitting state. For $LiLaP_4O_{12}$, LaP_3O_9 and $ScPO_4$, the $\Delta E(Bi^{3+})$ value is below 1.39 eV which suggests an emission of A character, *i.e.* the energy absorbed by the C and/or MMCT state(s) is relaxed non radiatively to the lowest-lying A state. The resulting "true" Stokes shifts would then amount 0.9, 1.15 and 1.18 eV for $LiLaP_4O_{12}$, LaP_3O_9 and $ScPO_4$ respectively. The situation in the zircon phases YPO_4 and $LuPO_4$ remains puzzling for two reasons: (1) they show an emission below 250 nm that is not observed in the other phosphates. The corresponding energy is beyond that of the A state and thus, the emission cannot be ascribed to an A transition; (2) the lower-lying UV emission was ascribed to CT processes in earlier reports[30,176] but none of them reproduce the UV excitation within ±0.35 eV (see also Fig. 5). At this point, we are inclined to suggest an alternative interpretation of these data in which the lowest UV emission would be of A character, with a Stokes shift of ≈0.2 eV that accounts for a small electron–phonon coupling. The higher-lying UV emission is tentatively ascribed to the C state ($^1P_1 \rightarrow {}^1S_0$ transition) with comparatively higher intensity because the transition is spin allowed. The corresponding Stokes shift would amount to 0.35 eV, again confirming a small electron–phonon coupling. We are aware that emission from the C state is not usually reported in the archival literature, but we note that Blasse *et al.* have already considered this as a possible option in the case of Bi^{3+}-activated oxychlorides.[1] Assigning the 5.46–5.51 eV excitation band to a C transition requires to reconsider the assignment of the excitation bands that appear in the 150–200 nm spectral region and that are related to Bi doping.[227] For this, we consider the results of DFT calculations that are reported in the data sheet MP-5132 of YPO_4[70] based on the calculation methodology detailed in ref. 228. These calculations reveal several DOS maxima at 5.93, 6.26, 6.60, 6.94, 7.44, 8.11 and 8.50 eV for a bandgap estimated at 5.80 eV. All these signals pertain to empty Y-3d orbitals and constitute most of the lowest CB states. As it is usual for DFT calculations, these energies are downshifted with respect to experimental data and must be rescaled. For this, we can use the Vacuum Referred Binding Energy (VRBE) diagram of YPO_4 published in ref. 227 in which the energy of the valence top (E_V), the host exciton (E^X) and the conduction band bottom (E_C) are determined at −9.77, −1.22 and −0.53 eV, respectively. On these grounds, Awater *et al.* have located the $^1S_0$ ground state of Bi^{3+} at −8.2 eV, *i.e.* 7.3 eV below the energy $(E_C + E^X)/2$, where 7.3 eV is the energy that was considered for Bi → Y MMCT. We did not obtain such a value with eqn (4) but a value closer to 4.95 eV. Using this latter, we locate the $^1S_0$ ground state at −5.82 eV, the $^3P_1$ state at −1.89 eV and the $^1P_1$ state at −0.38 eV, *i.e.* in resonance with the CBM. The other states constituting the CB can be obtained by assuming that the calculated bandgap of 5.80 eV coincides with E_C. In this frame, we find that transitions from $^1S_0$ to these states fairly reproduce the excitation features that are observed in the 150–200 nm spectral range.

Table 6 Spectroscopic properties of some Bi^{3+}-doped phosphates. Energies in eV

Compound	$E_{exp}(Bi^{3+})$	$E_{em}(Bi^{3+})$	$E_A(Bi^{3+})$	$E_c(Bi^{3+})$	$E_{CT}(Bi^{3+})$
YPO_4 (Y[8], Y[8]) (↓)	5.46–5.51	5.08–5.10	3.93	5.45	4.95
YPO_4 (Y[8], Bi[8]) – pairs		3.70–3.75			4.44
$LuPO_4$ (Lu[8], Lu[8]) (↓)	5.51	5.25	3.95	5.49	4.79
$LuPO_4$ (Lu[8], Bi[8]) – pairs		3.70			4.44
$ScPO_4$ (Sc[8], Sc[8]) (↓)	4.96	2.82	4.00	5.60	4.79
$LaPO_4$ (La[9], La[9]) (↓)	5.17	2.76	4.15	5.87	5.07
$LiLaP_4O_{12}$ (La[8], La[8]) (↓)	5.27–5.40	3.02–2.95	3.87	5.32	5.52
LaP_3O_9 (La[8], La[8]) (↓)	5.27	2.72	3.87	5.32	5.13

5.1.2. The borates with calcite crystal structure. The spectroscopic data of $LuBO_3$:Bi and $ScBO_3$:Bi in their calcite form are not reproduced by eqn (18). Those of the isostructural $InBO_3$:Bi and $GaBO_3$:Bi are reproduced based on Bi → In and Bi → Ga MMCT, but their associated $\Delta E(Bi^{3+})$ values suggest an emission from the A state. The properties of these borates are compared in Table 7. The position of the C state is calculated from eqn (19). We find a good correspondence between $E_{exp}(Bi^{3+})$ and $E_C(Bi^{3+})$ that motivates a re-interpretation of the spectral data. For this, we plot in Fig. 7 the energy level structure of Bi^{3+} in these compounds.

The absorption of $ScBO_3$:Bi and $LuBO_3$:Bi fits well a C transition but the energy of the UV emission in these compounds is beyond that of the A state, as in the case of YPO_4:Bi and $LuPO_4$:Bi. Therefore, this emission cannot be ascribed to the A state. The Bi → Sc and Bi → Lu MMCT states are not excited in direct using 4.32–4.35 eV photons and should not interfere with the emission process unless they are populated by thermal motion. The absorption of $InBO_3$:Bi and $GaBO_3$:Bi is consistent with MMCT and C transitions. This offers two possible relaxation paths before emission. The $\Delta E(Bi^{3+})$ values given in Table 5 suggest an emission from the A state after radiationless relaxation. In this frame, the corresponding "true" Stokes shift would amount 0.56 eV for $GaBO_3$:Bi and 0.30 eV for $InBO_3$:Bi. As for $ScBO_3$:Bi and $LuBO_3$:Bi, our model does not support an attribution of the UV emission of $GaBO_3$:Bi to an A transition.

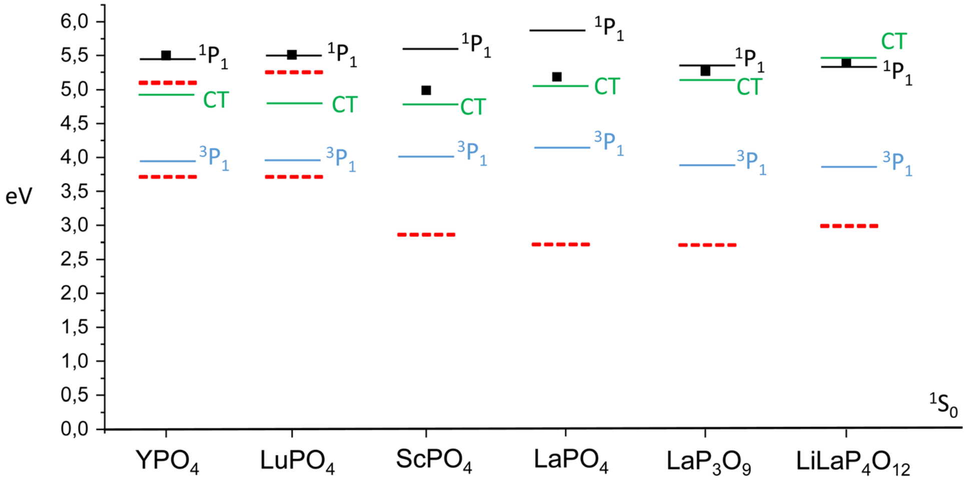


Fig. 6 Energy level structure of Bi^{3+} in some phosphates. Solid lines denote the calculated position of the C ($^1P_1$), A ($^3P_1$) and CT states and dotted lines denote the emission energy. ■ denotes the experimental excitation.

Regarding the concomitance of C and MMCT transitions, it is noted that the 4.60 and 4.71 eV excitation signals of Sc_2O_3:Bi and Y_2O_3:Bi can be interpreted either as resulting from Bi → Sc or Bi → Y MMCT (4.45 and 4.73 eV, respectively) or from absorption in the C state (4.62 and 4.42 eV, respectively). The failure of our model for Bi^{3+}-doped alkali halides, Sr_3AlO_4F, $Ba_2ZnGe_2O_7$ and $YAl_3B_4O_{12}$ is, to date, unexplained. Perhaps the Bi ions do not occupy a regular lattice site in these compounds, for instance due to local charge compensation, or/and the spectral data are not Bi-related, as for instance in La_2MoO_6:Bi and $La_2Mo_2O_9$:Bi.[229]

5.2. The compounds reproduced with an assignment that differs from the literature

In general, these compounds should be regarded as CT-absorbers rather than A-absorbers. This is for instance the case of the gallate $CaGa_2O_4$, the tungstate La_3BWO_9 and the tantalates $CaLaMgTaO_6$ and $ALaTa_2O_7$ (A = Na, K, Rb). For all, the ΔE value exceeds 1.41 eV, so these compounds are also perceived as CT-emitters. The scheelites $CaMoO_4$:Bi and $CaWO_4$:Bi are also regarded as CT emitters. In these compounds, the excitation signals in the 4.32–4.40 eV spectral range are well reproduced by Bi(Ca) → W and Bi(Ca) → Mo MMCT transitions (4.29 and 4.23 eV, respectively).

Other situations are examined below in more details.

5.2.1. Germanates. We focus here on germanates incorporating tetrahedrally coordinated Ge^{4+} ions. 12 compounds of that type (disregarding $Ba_2ZnGe_2O_7$) are contained in Table 5. All of them are ascribed as A-absorbers/A-emitters and half are confirmed to be so. The remaining ones are compiled in Table 8 where the data pertain to Bi^{3+} doping in trivalent cation sites.

Owing to their low $\Delta E(Bi^{3+})$ values, these germanates present an emission from the A state but they absorb in the CT state (or/and possibly the C state in the case of $LiScGeO_4$ (signal at 4.43 eV), $NaGdGeO_4$ and Lu_2GeO_5(Lu2 site)). The A state is located systematically below the absorbing state in these compounds. We assume that it is populated after internal radiationless relaxation.

5.2.2. Perovskites. We concentrate here on oxidic perovskites. Their data are compiled in Table 9. Bi^{3+} doping was considered in all possible cation sites.

We first concentrate on ''simple'' perovskites. The compounds in bold are viewed as CT absorbers and as CT emitters when $\Delta E(Bi^{3+}) > 1.41$ eV. The spectroscopic properties of Bi^{3+} in perovskites have recently been re-inspected by Srivastava

Table 7 Spectroscopic properties of some Bi^{3+}-doped borates with calcite structure

Compound	$E_{exp}(Bi^{3+})$	$E_{em}(Bi^{3+})$	$E_A(Bi^{3+})$	$E_C(Bi^{3+})$	$E_{CT}(Bi^{3+})$
$GaBO_3$ (Ga[6], Ga[6]) (↓)	4.44	2.93	3.49	4.45	4.53
	4.49	4.28			
$ScBO_3$ (Sc[6], Sc[6]) (↓)	4.32	4.10	3.44	4.42	4.79
$InBO_3$ (In[6], In[6]) (↓)	4.37–4.43	3.10	3.40	4.35	4.67
$LuBO_3$ (Lu[6], Lu[6]) (↓)	4.35	4.08	3.39	4.32	4.79

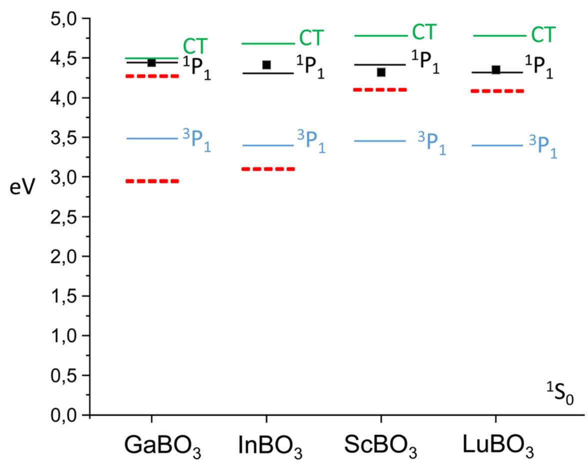


Fig. 7 Energy level structure of Bi^{3+} in borates with calcite structure. Solid lines denote the calculated position of the C ($^1P_1$), A ($^3P_1$) and CT states and dotted lines denote the emission energy. ■ denotes the experimental excitation.

Table 8 Spectroscopic properties of some Bi^{3+}-doped germanates based on tetrahedrally-coordinated Ge^{4+} ions

Compound	$E_{exp}(Bi^{3+})$	$E_A(Bi^{3+})$	$E_C(Bi^{3+})$	$E_{CT}(Bi^{3+})$	$\Delta E(Bi^{3+})$	Emission
$LiScGeO_4$ (Sc[6], Ge[4]) (↓)	4.13, 4.43	3.55	4.68	4.23	0.05	A
$LiLuGeO_4$ (Lu[6], Ge[4]) (↓)	4.06	3.50	4.57	4.29	0.27	
$NaYGeO_4$ (Y[6], Ge[4]) (↓)	4.09	3.47	4.49	4.42	0.12	
$NaGdGeO_4$ (Gd[6], Ge[4]) (↓)	4.10	3.42	4.39	4.42	0.15	
$NaLuGeO_4$ (Lu[6], Ge[4]) (↓)	4.10	3.48	4.53	4.36	0.18	
Lu_2GeO_5 (Lu1[6], Ge[4]) (↓)	4.35	3.58	4.73	4.55	0.58	
Lu_2GeO_5 (Lu2[6], Ge[4]) (↓)		3.44	4.42	4.68	0.85	

et al.[137] In particular, it was shown that $SrZrO_3$:Bi should be considered as a A emitter rather than a CT emitter, as ascribed in an earlier report.[138] This new insight was of great help to fix the A ↔ MMCT threshold value for $\Delta E(Bi^{3+})$. The A character of the Bi^{3+} emission in $CaZrO_3$ is also confirmed. The recent work from Back *et al.* on this system demonstrated that the lowest-lying excitation band consists of 2 components at 4.15 and 3.89 eV that were assigned respectively to Bi(Ca)→ Zr MMCT and A transitions.[125] Our calculations reproduce these data within ±0.35 eV but suggest an opposite order for the A and CT state. In calcium and strontium stannates, two emissions were observed in correspondence with a unique excitation signal.[124,138] These emissions were ascribed to A and CT states. We confirm these assignments. Note that the lowest-lying excitation band of $CaSnO_3$:Bi was shown to consist, in fact, of 2 components at 4.22 and 4.03 eV that were assigned respectively to Bi(Ca)→ Sn MMCT and A transitions.[125] Again, our calculations reproduce these data within ±0.35 eV but suggest an opposite order for the A and CT state.

The underlined perovskites are confirmed as A-absorbing and A-emitting, in agreement with the literature. Here, the CT state is well above the A state and does interfere with the relaxation process. $LaGaO_3$:Bi is an exception. Its excitation spectrum (Fig. 1 of ref. 205) was digitized (after conversion in eV) and reproduced using a sum of 2 Gaussian functions, featuring maxima at 3.87 eV

Table 9 Spectroscopic properties of some Bi^{3+}-doped perovskites. The compounds are listed by increasing values of $he(Bi^{3+})$

Compound	$E_{exp}(Bi^{3+})$	$E_A(Bi^{3+})$	$E_{CT}(Bi^{3+})$	$\Delta E(Bi^{3+})$	Emission
Simple perovskites					
$LaAlO_3$ (La[12], La[12]) (↑)	4.35–**4.56**	5.17	4.90	1.29	A
$NaNbO_3$ (Na1[9], Nb[6]) (↑)	**3.78**	5.10	3.72	2.99	CT
$NaNbO_3$ (Na2[8], Nb[6]) (↑)		4.75	3.66	2.70	CT
$CaHfO_3$ (Ca[8], Hf[6]) (↑)	**4.03**	4.44	3.89	1.34	A
$CaZrO_3$ (Ca[8], Zr[6]) (↑)	**3.92–4.02**	4.43	4.05	1.20	A
$CaSnO_3$ (Ca[8], Sn[6]) (↑)	**4.03–4.07**	4.43	3.83	1.23	A
				1.93	CT
$CaTiO_3$ (Ca[8], Ti[6]) (↑)	*3.35*, **3.65–3.87**	4.41	3.77	2.28	CT
$SrZrO_3$ (Sr[8],Zr[6]) (≈)	**4.09**	4.36	4.17	1.41	A
$SrSnO_3$ (Sr[8],Sn[6]) (↑)	**4.13**	4.33	3.94	1.32	A
				1.67	CT
$YAlO_3$(Y[8],Y[8]) (↓)	4.43	4.16	4.79	0.04	A
$GdAlO_3$(Gd[8],—)	4.24	4.12			A
$LaYO_3$(La[8],Y[6]) (↓)	3.76	4.11	4.62	0.75	A
$MgGeO_3$(Mg1[6],Ge1[4]) (↓)	4.27	3.92	4.74	0.01	A
$MgGeO_3$(Mg1[6],Ge2[4]) (↓)			4.94	0.21	
$LaGaO_3$ (La[7],Ga[6]) (↓)	**4.02–4.03**	3.75	4.11	0.37	A
$LaInO_3$(In[6],La[8]) (↓)	3.65	3.47	4.34	0.19	A
$LaScO_3$(Sc[6],La[8]) (↓)	3.30–3.71,**4.54**	3.45	4.67	<1.07	A
Double perovskites					
Sr_2YNbO_6 (Sr[8], Nb[6]) (↑)	*3.31,* **3.63–3.77**	4.39	3.72	1.86	CT
Ca_2MgWO_6 (Ca[8], W[6]) (↑)	**3.67**	4.43	3.77	2.07	CT
La_2MgGeO_6(La1[9],Ge[6]) (↑)	4.13–4.24	4.40	3.72	1.71	CT
La_2MgGeO_6(La2[9],Ge[6]) (↑)		4.40	3.72	1.71	CT
Ba_2GdNbO_6 (Ba[8], Nb[6]) (↓)	*3.28,* **3.93**	4.39	3.89	1.76	
$CaLaMgTaO_6$ (La/Ca[8], Ta[6]) (↑)	**3.73**	4.26	3.55	1.64	CT
La_2ZnTiO_6 (La[8], Ti[6]) (≈)	**3.54**	4.12	3.83	0.79	A
La_2MgTiO_6 (La[8], Ti[6]) (≈)	**3.64**	4.12	3.89	1.00	A
La_2LiSbO_6(La[8],La[8]) (↓)	3.96	4.12	5.01	0.03	A
La_2LiNbO_6 (La[8],Nb[6]) (↑)	**3.76**	4.11	3.72	1.33	A
Ca_2MgWO_6 (Mg[6],W[6]) (≈)	**3.67**	3.80	3.77	1.44	CT
$CaLaMgTaO_6$ (Mg[6],Ta[6]) (≈)	**3.73**	3.77	3.55	1.15	A
La_2ZnTiO_6 (Zn[6], Ti[6]) (↓)	**3.54**	3.82	4.45	0.13	A
La_2MgTiO_6(Mg[6],Ti[6])(↓)	**3.64**	3.75	5.10	0.07	A
La_2MgGeO_6 (Mg[6], Ge[6]) (↓)	**4.13–4.24**	3.73	4.22	<0.73	A
Sr_2YNbO_6(Y[6],Nb[6]) (↓)	3.31,3.63–*3.77*	3.30	4.34	<0.68	A
Ba_2GdNbO_6(Gd[6],Nb[6])(↓)	3.28, *3.93*	3.34	4.39	0.44	A

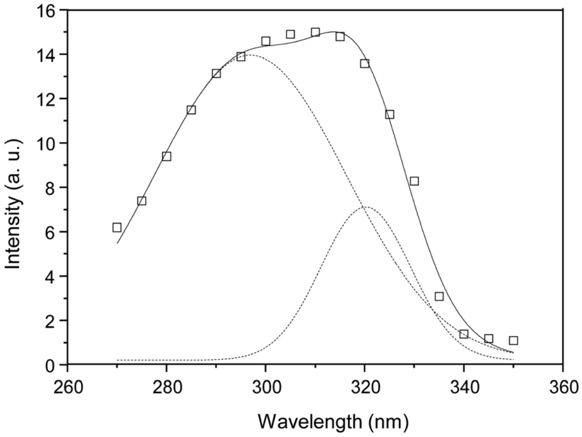


Fig. 8 Spectral decomposition of the excitation spectrum of $LaGaO_3$:Bi after digitization of Fig. 1 in ref. 205.

(320 nm) and 4.17 eV (297 nm), respectively, see Fig. 8. This matches well the calculated position of the A (3.75 eV) and CT states (4.11 eV). We conclude that the excitation of Bi^{3+} in this compound involves both transitions but emission occurs from the lowest-lying A state. Note that the C transition is calculated at 5.08 eV (244 nm) by means of eqn (19) using $E_A(Bi^{3+})$ = 3.75 eV. This matches well the position of the short wavelength excitation band ascribed earlier to $^1S_0 \rightarrow {}^1P_1$ transition.[204]

At last, it is noted that the A absorption in $LaAlO_3$, $LaScO_3$ and $LaInO_3$ is not reproduced by substituting Bi to La sites. This result needs further experimental checking.

We move to the double perovskites. Here, the spectral analysis depends a lot on the nature of the cation site that is considered for Bi^{3+}. We can make two categories whether the compound contains or not a trivalent cation site available for Bi^{3+}. Let's start with the most intuitive situation involving doping in trivalent cation sites, *i.e.* La^{3+}, Gd^{3+} and Y^{3+} in the present cases. Here, we observe that the lowest-lying state (A or CT) is always the emitting one if the energy separation from the other state (CT or A) is large enough. In $La_2ZnTiO_6$and La_2MgTiO_6, the A and CT states are resonant and the emission is of A type, in agreement with earlier reports.[165,167] Let us now consider Bi^{3+} doping in divalent cation sites, as it occurs *e.g.* in Ca_2MgWO_6. In this compound, Bi^{3+} absorbs and emits from the CT state, whatever the doping site. In La_2MgGeO_6, La_2MgTiO_6, La_2ZnTiO_6 and $CaLaMgTaO_6$, a hypothetic doping in the Mg^{2+} or Zn^{2+} site would cause a depression of the A state at the level or even below the CT state, with resulting emission of A type. This doping, however, is not favored by the difference in charge and ionic radius with respect to Bi^{3+} and, to our opinion, not probable.

We conclude this review by analyzing the recently reported cases of Sr_2YNbO_6:Bi[117] and Ba_2GdNbO_6:Bi.[145] Ba_2GdNbO_6:Bi shows one emission at 2.67 eV ascribed to A transition. This emission is excited at 3.28 eV (ascribed to A transition) and 3.93 eV (ascribed to C transition).[145] The 3.28 eV excitation is confirmed of A character for Bi^{3+} located in the Gd^{3+} site. The corresponding C transition is calculated at 4.21 eV by means of eqn (19), which is consistent with the 3.93 eV signal within ±0.35 eV. We note that this signal would also be well reproduced by a Bi(Ba) → Nb MMCT, considering Bi doping in the Ba^{2+} site (see Table 9). The A character of the emission is confirmed. Sr_2YNbO_6:Bi shows two emission bands that were ascribed to A transitions from Bi^{3+} loaded in Y and Sr sites, in correspondence with excitation signals at 3.31 eV (La site) and 3.63–3.77 eV (Sr site), respectively.[117] Our calculations nicely reproduce the A absorption (and subsequent A emission) for Bi doping in the Y site, but not for Bi doping in the Sr sites. In this case, our calculations rather suggest a Bi(Sr) → Nb MMCT excitation followed by CT emission.

6. Conclusion

The luminescence properties of Bi^{3+} in 177 inorganic compounds are reviewed and rationalized based on luminescence–structure relationships. Two semi-empirical models are used, one involving the electronegativity of Bi^{3+} and its closest d^0 or d^{10} cation neighbor (when present) and the second one involving the nephelauxetic effect at the crystal site occupied by Bi^{3+} in the host lattice. In this concern, the calculation of the nephelauxetic function, *he*, is described step-by-step for an easy implementation in a home-made calculator. Examples are given in the ESI.† These models allow an easy and rapid estimation of the energy associated with the intra-ionic $^1S_0(6s^2) \rightarrow {}^3P_1(6s^1p^1)$ transition (E_A) and with the extra-ionic Bi^{3+}-to-metal charge transfer (E_{CT}) within an accuracy of ±0.35 eV in any kind of compound if its crystal structure is known. A criterion based on E_A, E_{CT} and the experimental value of the Stokes shift is introduced to help in fixing the origin of the Bi^{3+}-related emission. It is hoped that these methods will be widely used in a routine manner, *e.g.* in the form of a machine-learning algorithm, to guide the design of new compounds that incorporate Bi^{3+} as sensitizer or as a luminescent activator. The perspectives, in this concern, are real, as demonstrated by the increasing desire for Bi^{3+}-doped compounds applicable in *e.g.* lighting,[230–234] persistent luminescence,[235–238] anti-counterfeiting[239–242] or optical thermometry.[243–249]

Conflicts of interest

There are no conflicts to declare.

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There is no competing funding to declare.

References

1 G. Blasse and A. Bril, Investigations on Bi^{3+}-activated phosphors, *J. Chem. Phys.*, 1968, **48**, 217.
2 G. Boulon, F. Gaume-Mahn, C. Pedrini, B. Jacquier, J. Janin and D. Curie, *Luminescence processes in Bi^{3+} centers, in: Luminescence of Crystals, Molecules and Solutions*, ed. F. Williams *et al.*, Plenum Press, New York, 1973.

3 P. Boutinaud and E. Cavalli, Predicting metal-to-metal charge transfer in closed shell transition metal oxides doped with Bi^{3+}, *Chem. Phys. Lett.*, 2011, **503**, 239.
4 P. Boutinaud, Revisiting the spectroscopy of the Bi^{3+} ion in oxide compounds, *Inorg. Chem.*, 2013, **52**, 6028.
5 M. Amer and P. Boutinaud, On the character of the optical transitions in closed-shell transition metal oxides doped with Bi^{3+}, *Phys. Chem. Chem. Phys.*, 2017, **19**, 2591.
6 R. H. Awater and P. Dorenbos, The Bi^{3+} 6s and 6p electron binding energies in relation to the chemical environment of inorganic compounds, *J. Lumin.*, 2017, **184**, 221.
7 R. H. Awater and P. Dorenbos, Towards a general concentration quenching model of Bi^{3+} luminescence, *J. Lumin.*, 2017, **188**, 487.
8 P. Boutinaud, On the luminescence of Bi^{3+} pairs in oxidic compounds, *J. Lumin.*, 2018, **197**, 228.
9 J. P. Xue, X. F. Wang, J. H. Jeong and X. H. Yan, Spectral and energy transfer in Bi^{3+}-Re^{n+} (n = 2,3,4) co-doped phosphors: extended optical applications, *Phys. Chem. Chem. Phys.*, 2018, **20**, 11516.
10 H. C. Swart and R. E. Kroon, Ultraviolet and visible luminescence from bismuth doped materials, *Opt. Mater.: X*, 2019, **2**, 100025.
11 A. Krasnikov, E. Mihokova, M. Nikl, S. Zazubovich and Ya Zhydachevskyy, Luminescence Spectroscopy and Origin of Luminescence Centers in Bi-Doped Materials, *Crystals*, 2020, **10**, 208.
12 P. Boutinaud, On the spectroscopy of Bi^{3+} in d^{10} post-transition metal oxides, *J. Lumin.*, 2020, **223**, 117219.
13 P. Dang, D. Liu, G. Li, A. A. Al Kheraif and J. Lin, Recent advances in bismuth ion-doped phosphor materials: structure design, tunable photoluminescence properties and application in white LEDS, *Adv. Opt. Mater.*, 2020, **8**, 1901993.
14 P. Boutinaud and M. Brik, The optical properties of Bi^{3+} and Sb^{3+} in $YNbTiO_6$ analyzed by means of DOS and semi-empirical calculations, *J. Mater. Chem. C*, 2020, **8**, 2086.
15 P. Dorenbos, Electronic structure of Bi-activated luminescent compounds and pure bismuth photocatalytic compounds, *ECS J. Solid State Sci. Technol.*, 2021, **10**, 086002.
16 J. A. Duffy and M. D. Ingram, New correlation between s-p spectra and the nephelauxetic ratio β: applications in molten salt and glass chemistry, *J. Chem. Phys.*, 1971, **54**, 443.
17 G. Blasse, The ultraviolet absorption bands of Bi^{3+} and Eu^{3+} in oxides, *J. Solid State Chem.*, 1972, **4**, 52.
18 G. Blasse, A. W. M. Braam and M. Heerschop, Influence of crystal structure on the luminescence of ions with s^2 configuration, *J. Solid State Chem.*, 1977, **20**, 63.
19 J. A. Duffy, Prediction of ultraviolet absorption spectra of p-block ions in oxide media. Possible application to redox equilibria in glasses and molten salts, *J. Chem. Soc., Faraday Trans.*, 1978, **2**, 1504.
20 G. Blasse and A. C. van der Steen, Luminescence characteristics of Bi^{3+}-activated oxides, *Solid State Commun.*, 1979, **31**, 993.
21 L. Wang, Q. Sun, Q. Liu and J. Shi, Investigation and application of quantitative relationship between sp energy levels of Bi^{3+} ion and host lattice, *J. Solid State Chem.*, 2012, **191**, 142.
22 A. Wolfert and G. Blasse, Luminescence of the Bi^{3+} ion in compounds LnOCl (Ln = La,Y,Gd), *Mater. Res. Bull.*, 1984, **19**, 67.
23 A. Wolfert, E. W. J. L. Oomens and G. Blasse, Host lattice dependence of the Bi^{3+} luminescence in orthoborates $LnBO_3$ (with Ln = Sc,Y,La,Gd or Lu), *J. Solid State Chem.*, 1985, **59**, 280.
24 H. Zhiran and G. Blasse, Energy transfer phenomena in luminescent materials based on GdB_3O_6, *Mater. Phys. Chem.*, 1985, **12**, 257.
25 A. M. Van de Craats and G. Blasse, The quenching of bismuth (III) luminescence in yttrium oxide (Y_2O_3), *Chem. Phys. Lett.*, 1995, **243**, 559.
26 A. M. Srivastava, On the luminescence of Bi^{3+} in the pyrochlore $Y_2Sn_2O_7$, *Mater. Res. Bull.*, 2002, **37**, 745.
27 A. A. Setlur and A. M. Srivastava, The nature of Bi^{3+} luminescence in garnet hosts, *Opt. Mater.*, 2006, **29**, 410.
28 Y. Zorenko, V. Gorbenko, T. Voznyak, V. Vitovsky, S. Nedilko and M. Nikl, Luminescence of Bi^{3+} ions in $Y_3Al_5O_{12}$:Bi single crystalline films, *Radiat. Meas.*, 2007, **42**, 882.
29 Y. Zorenko, V. Gorbenko, T. Voznyak, V. Jary and M. Nikl, Luminescence spectroscopy of the Bi^{3+} single and dimer centers in $Y_3Al_5O_{12}$:Bi single crystalline films, *J. Lumin.*, 2010, **130**, 1963.
30 A. M. Srivastava and S. J. Camardello, Concentration dependence of the Bi^{3+} luminescence in $LnPO_4$ (Ln = Y^{3+}, Lu^{3+}), *Opt. Mater.*, 2015, **39**, 130.
31 M. Srivastava and H. A. Comanzo, The ultraviolet and visible luminescence of Bi^{3+} in the orthorhombic perovskite, $GdAlO_3$, *Opt. Mater.*, 2017, **63**, 118.
32 X. Wang, P. Boutinaud, L. Li, J. Cao, P. Xiong, X. Li, H. Luo and M. Peng, Novel persistent and tribo-luminescence from bismuth ion pairs doped strontium gallate, *J. Mater. Chem. C*, 2018, **6**, 10367.
33 P. Boutinaud, Spectroscopic investigations of calcium fluoroapatites doped with Bi^{3+}, *J. Lumin.*, 2019, **205**, 237.
34 R. Moncorgé and G. Boulon, Investigation of the absorption and emission properties along with energy transfer in pure and Bi^{3+}-doped YVO_4, *J. Lumin.*, 1979, **18/19**, 376.
35 M. Ilmer, B. Grabmaier and G. Blasse, Luminescence of Bi^{3+} in gallate garnets, *Chem. Mater.*, 1994, **6**, 204.
36 G. Blasse, C. de Mello Donega, I. V. Berezovskaya and V. P. Dotsenko, The luminescence of bismuth (III) in indium borate, *Solid State Commun.*, 1994, **91**, 29.
37 A. M. Srivastava and W. W. Beers, On the impurity trapped exciton luminescence in $La_2Zr_2O_7$:Bi^{3+}, *J. Lumin.*, 1999, **81**, 293.
38 V. Babin, V. Gorbenko, A. Krasnikov, A. Makhov, M. Nikl, S. Zazubovich and Yu Zorenko, Photoluminescence of $Lu_3Al_5O_{12}$:Bi and $Y_3Al_5O_{12}$:Bi single crystalline films, *Radiat. Meas.*, 2010, **45**, 331.

39 A. M. Srivastava, H. A. Comanzo and S. J. Camaradello, On the "Bi^{3+}-Ti^{4+}" charge transfer transition in the pyrochlore $Y_2Ti_2O_7$:Bi^{3+}, *Opt. Mater.*, 2015, **48**, 31.
40 V. Babin, K. Chernenko, P. Demchenko, E. Mihokova, M. Nikl, I. Pashuk, T. Shalapska, A. Voloshinvskii and S. Zazubovich, Luminescence and excited state dynamics in Bi^{3+}-doped $LiLaP_4O_{12}$ phosphates, *J. Lumin.*, 2016, **176**, 324.
41 S. Mahlik, M. Amer and P. Boutinaud, Energy Level Structure of Bi^{3+} in Zircon and Scheelite Polymorphs of YVO_4, *J. Phys. Chem. C*, 2016, **120**, 8261.
42 V. Babin, V. Gorbenko, A. Krasnikov, A. Makhov, E. Mihokova, M. Nikl, S. Zazubovich and Yu Zorenko, Origin of Bi^{3+}-related luminescence centres in $Lu_3Al_5O_{12}$:Bi and $Y_3Al_5O_{12}$:Bi single crystalline films and the structure of the of their relaxed excited states, *Phys. Status Solidi B*, 2012, **249**, 1039.
43 M. Buryi, P. Bohacek, K. Chernenko, A. Krasnikov, V. V. Laguta, E. Mihokova, M. Nikl and S. Zazubovich, Luminescence and photo-thermally stimulated defect-creation processes in Bi^{3+}-doped single crystals of lead tungstate, *Phys. Status Solidi B*, 2016, **253**, 895.
44 V. Tsiumra, A. Zhyshkovych, T. Malyi, Y. Chornodolskyy, V. Vistovskyy, S. Syrotyuk, Ya Zhydachevskyy, A. Suchocki and A. Voloshinovskii, Localized exciton luminescence in YVO_4:Bi^{3+}, *Opt. Mater.*, 2019, **89**, 480.
45 A. Krasnikov, V. Tsiumra, L. Vasylechko, S. Zazubovich and Ya Zhydachevskyy, Photoluminescence origin in Bi^{3+}-doped YVO_4, $LuVO_4$ and $GdVO_4$, *J. Lumin.*, 2019, **212**, 52.
46 M. Baran, A. Kissabekova, A. Krasnikov, A. Lushchik, A. Suchocki, V. Tsiumra, L. Vasylechko, S. Zazubovich and Ya Zhydachevskyy, Exciton-like luminescence of Bi^{3+}-doped yttrium niobates, *Nucl. Instrum. Methods Phys. Res., Sect. B*, 2020, **463**, 7.
47 B. Lou, J. Wen, J. Cai, Y.-Y. Yeung, M. Yin and C.-K. Duan, First-principles study of Bi^{3+}-related luminescence and traps in the perovskites $CaMO_3$ (M = Zr, Sn, Ti), *Phys. Rev. B*, 2021, **103**, 075109.
48 P.-P. Sun, D. R. Kripalani, L. Bai and K. Zhou, Prediction of the role of bismuth dopants in organic-inorganic lead halide perovskites on photoelectric properties and photovoltaic performance, *J. Phys. Chem. C*, 2019, **123**, 12684.
49 Q. Chen, Z. Feng, M. Liu, B. Lou, C.-G. Ma and C.-K. Duan, The photoluminescence of isolated and paired Bi^{3+} ions in layered LnOCl crystals: a first-principles study, *Phys. Chem. Chem. Phys.*, 2022, **24**, 14064.
50 X. Wang, C. Heng, Z. Qiao and L. Ning, First-principles study on luminescent properties of Bi^{3+}-doped $ALuGeO_4$ (A = Li, Na): insights into effects of host cation on emission wavelength, *J. Lumin.*, 2022, **244**, 118700.
51 Z. Feng, B. Lou, M. Yin, Y.-Y. Yeung, H.-T. Sun and C.-K. Duan, First-principles study of Bi^{3+}-related luminescence and electron and hole traps in $(Y/Lu/La)PO_4$, *Inorg. Chem.*, 2021, **60**, 4434.
52 J. Liu, S. Guo, H. Wu, X. Zhang, J. Li and K. Zhou, Synergetic effects of Bi^{5+} and oxygen vacancies in bismuth(V)-rich Bi4O7 nanosheets for enhanced near-infrared light driven photocatalysis, *J. Mater. Sci. Technol.*, 2021, **85**, 1.
53 P. Boutinaud, Revisiting Duffy's model for Sb^{3+} and Bi^{3+} in double halide perovskites: emergence of a descriptor for machine learning, *Opt. Mater.: X*, 2021, **11**, 100082.
54 P. Boutinaud, ERRATUM: Revisiting Duffy's model for Sb^{3+} and Bi^{3+} in double halide perovskites: emergence of a descriptor for machine learning, *Opt. Mater.: X*, 2021, **12**, 100109.
55 C. E. Moore, *Atomic Energy Levels*, National Bureau of Standards, Washington DC, 1952.
56 A. J. Bruce and J. A. Duffy, Thin films absorption spectra of lower valent p block halides, *Thin Solid Films*, 1983, **101**, 179.
57 J. A. Duffy and M. D. Ingram, Use of thallium(I), lead(II), and bismuth(III) as spectroscopic probes for ionic–covalent interaction in glasses, *J. Chem. Phys.*, 1970, **52**, 3752.
58 J. C. Phillips, Dielectric definition of electronegativity, *Phys. Rev. Lett.*, 1968, **20**, 550.
59 J. A. Van Vechten, Quantum dielectric theory of electronegativity in covalent systems. I. Electronic dielectric constant, *Phys. Rev.*, 1969, **182**, 891.
60 J. A. Van Vechten, Quantum dielectric theory of electronegativity in covalent systems. II. Ionization potentials and interband transition energies, *Phys. Rev.*, 1969, **187**, 1007.
61 J. C. Phillips, Ionicity of the chemical bond in crystals, *Rev. Mod. Phys.*, 1970, **42**, 317.
62 B. F. Levine, d-electron effects on bond susceptibilities and ionicities, *Phys. Rev. B: Solid State*, 1973, **7**, 259.
63 J. S. Shi and S. Y. Zhang, Barycenter of energy of lanthanide $4f^{n-1}5d$ configuration in inorganic crystals, *J. Phys. Chem. B*, 2004, **108**, 18845.
64 Q. Sun, J. Wang and J. Shi, Prediction and assignment of site occupation and energy levels for Pb^{2+} ions in crystal hosts, *J. Solid State Chem.*, 2010, **183**, 1174.
65 P. Boutinaud, Luminescence of Sb^{3+} in closed shell transition metal oxides, *J. Lumin.*, 2019, **208**, 394.
66 K. Li and D. Xue, Estimation of Electronegativity Values of Elements in Different Valence States, *J. Phys. Chem. A*, 2006, **110**, 11332.
67 R. D. Shannon, Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides, *Acta Crystallogr., Sect. A: Cryst. Phys., Diffr., Theor. Gen. Crystallogr.*, 1976, **32**, 751.
68 K. Momma and F. Izumi, VESTA: a three-dimensional visualization system for electronic and structural analysis, *J. Appl. Crystallogr.*, 2008, **41**, 653.
69 N. E. Brese and M. O'Keefe, Bond-valence parameters for solids, *Acta Crystallogr., Sect. B: Struct. Sci.*, 1991, **47**, 192.
70 A. Jain, S. Ping Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder and K. A. Persson, Commentary: The Materials Project: A materials genome approach to accelerating materials innovation, *APL Mater.*, 2013, **1**, 011002.
71 Y. Wang, J. Ding, Y. Li, L. Yang, X. Ding and Y. Wang, A novel single-phase warm white emission phosphor

$Sr_3YAl_2O_{7.5}$:Bi^{3+}, Eu^{3+} with energy transfer from UV white LEDs, *RSC Adv.*, 2016, **6**, 42618.
72 J. Liao, P. Xie, W. Li, Z. Chen, Z. Liu, L. Chen, Z. Zheng and Y. Guo, White-light/tunable emissions in single-phased $BaLa_2Si_3O_{10}$: Eu^{3+}, Bi^{3+} phosphor for the simultaneous applications in white light-emitting diodes and luminous cement, *J. Mater. Sci.: Mater. Electron.*, 2020, **21**, 495.
73 H. Zhiran and G. Blasse, The luminescence of mercury-like ions in and the crystal structure of $SrLaBO_4$, *J. Solid State Chem.*, 1984, **55**, 23.
74 T. Lyu and P. Dorenbos, Vacuum-referred binding energies of bismuth and lanthanide levels in ARE(Si,Ge)O_4 (A = Li, Na; RE = Y; Lu): towards designing charge-carrier-trapping processes for energy storage, *Chem. Mater.*, 2020, **32**, 1192.
75 R. Sun, J. Zhao, J. Lu, S. Su, R. Liu, Z. Lei, C. Hu, B. Teng, J. Li, S. Sun and D. Zhong, Synthesis and characterizations of a color tunable single-phased warm white phosphor $Sr_3Lu(BO_3)_3$: Bi^{3+}, Eu^{3+}, *J. Lumin.*, 2023, **255**, 119570.
76 P. Shao, P. Xiong, Y. Xiao, Q. Chen, Y. Sun, N. Yan, D. Chen and Z. Yang, Novel spectral band: ultraviolet A mechanoluminescence from Bi^{3+}-doped $LiYGeO_4$, *J. Mater. Chem. C*, 2022, **10**, 16670.
77 Y. Wang, J. Ding and Y. Wang, Preparation and photoluminescence properties with the site-selected excitations of Bi^{3+} activated $Ba_3Sc_4O_9$ phosphors, *J. Am. Ceram. Soc.*, 2017, **6**, 2612.
78 H. Li, R. Pang, Y. Luo, H. Wu, S. Zhang, L. Jiang, D. Li, C. Li and H. Zhang, Structural micromodulation on Bi^{3+}-doped $Ba_2Ga_2GeO_7$ phosphor with considerable tunability of the defect-oriented optical properties, *ACS Appl. Electron. Mater.*, 2019, **1**(2), 229.
79 W. Chen, Z. Liu, L. Shen, C. Shen, L. Ding, Z. Zhang, H. Zhang, W. Xiang and X. Liang, Design and energy transfer mechanism for single-phased Gd_2MgTiO_6:Bi^{3+},Eu^{3+} tunable white light-emitting phosphors, *J. Mater. Sci.*, 2019, **54**, 4056.
80 S. K. Gupta, K. Sudarshan and R. M. Kadam, Tunable white light emitting $Sr_2V_2O_7$:Bi^{3+} phosphors: role of bismuth ion, *Mater. Des.*, 2017, **130**, 208.
81 K. Li, H. Lian, M. Shang and J. Lin, A novel greenish yellow-orange red $Ba_3Y_4O_9$:Bi^{3+}, Eu^{3+} phosphor with efficient energy transfer for UV-LEDs, *Dalton Trans.*, 2015, **44**, 20542.
82 H. Li, R. Pang, G. Liu, W. Sun, D. Li, L. Jiang, S. Zhang, C. H. Li, J. Feng and H. Zhang, Synthesis and luminescence properties of Bi^{3+}-activated K_2MgGeO_4: a promising high-brightness orange-emitting phosphor for WLEDs conversion, *Inorg. Chem.*, 2018, **57**, 12303.
83 V. Babin, L. Lipinska, E. Mihokova, M. Nikl, T. Shalapska, A. Suchocki, S. Zazubovich and Y. Zhydachevskyy, Time-resolved spectroscopy of Bi^{3+} centers in $Y_4Al_2O_9$, *Opt. Mater.*, 2015, **46**, 104.
84 G. Annadurai, L. Sun, H. Guo and X. Huang, Bright tunable white-light emissions from Bi^{3+}/Eu^{3+} co-doped phosphors *via* energy transfer for UV-excited white light-emitting diodes, *J. Lumin.*, 2020, **226**, 117474.
85 C. H. Kim, H. L. Park and S.-I. Mho, Photoluminescence of Eu^{3+} and Bi^{3+} in $Na_3YSi_3O_9$, *Solid State Commun.*, 1997, **101**, 109.
86 Z. Zhou, Y. Zhong, M. Xia, N. Zhou, B. Lei, J. Wang and F. Wu, Tunable dual emission of $Ca_3Al_4ZnO_{10}$:Bi^{3+}, Mn^{4+} *via* energy transfer for indoor plant growth lighting, *J. Mater. Chem. C*, 2018, **6**, 8914.
87 S. Lai, Z. Yang, J. Liao, J. Qiu, Z. Song, Y. Yang and D. Zhou, Investigation of persistent luminescence property of Bi^{3+}, Dy^{3+} co-doped $CdSiO_3$ phosphor, *Mater. Res. Bull.*, 2014, **60**, 714.
88 P. Dang, S. Liang, G. Li, H. Lian, M. Shang and J. Lin, Broad color tuning of Bi^{3+}/Eu^{3+}-doped $(Ba,Sr)_3Sc_4O_9$ solid solution compounds *via* crystal field modulation and energy transfer, *J. Mater. Chem. C*, 2018, **6**, 9990.
89 R. Jagannathan, S. P. Manoharan, R. P. Rao and T. R. N. Kutty, Luminescence and energy levels of Mn^{2+} in $LnMB_5O_{10}$ (Ln = La, Gd and Y; M = Mg, Zn, and Cd), *Jpn. J. Appl. Phys.*, 1990, **29**, 1991.
90 H. P. Awater and P. Dorenbos, X-ray induced valence change and vacuum referred binding energies of Bi^{3+} and Bi^{2+} in $Li_2BaP_2O_7$, *J. Phys. Chem. C*, 2016, **120**, 15114.
91 E. W. J. L. Oomen and G. Blasse, Luminescence of Bi^{3+} in the metaphosphates $Ln_3P_3O_9$ (Ln = Sc, Lu, Y, Gd, La), *J. Solid State Chem.*, 1988, **75**, 201.
92 S. Zhang, Y. Hu, R. Chen, X. Wang and Z. Wang, Photoluminescence and persistent luminescence in Bi^{3+}-doped Zn_2GeO_4 phosphors, *Opt. Mater.*, 2014, **36**, 1830.
93 P. Dorenbos, Lanthanide 4f-electron binding energies and the nephelauxetic effect in wide band gap compounds, *J. Lumin.*, 2013, **136**, 122.
94 R. Liu, W. Zhang, W. Liu and G. Li, Synthesis of a Bi^{3+}-doped Cs_2HfCl_6 double perovskite with highly efficient blue light emission at room temperature, *Inorg. Chem.*, 2021, **60**, 10451.
95 K. Chong, T. Hirai, T. Kawai, S. Hashimoto and N. Ohno, Optical properties of Bi^{3+} ions doped in $NaYF_4$, *J. Lumin.*, 2007, **122–123**, 149.
96 G. Xiong, L. Yuan, Y. Jin, H. Wu, Z. Li, B. Qu, G. Ju, L. Chen, S. Yang and Y. Hu, Aliovalent Doping and Surface Grafting Enable Efficient and Stable Lead-Free Blue-Emitting Perovskite Derivative, *Adv. Opt. Mater.*, 2020, **8**, 2000779.
97 K. P. Oboth, F. J. Lohmeier and F. Fischer, VUV and UV spectroscopy of Pb^{2+} and Bi^{3+} centres in alkaline-earth fluorides, *Phys. Status Solidi B*, 1989, **154**, 789.
98 M. Cai, T. Lang, T. Han, D. Valiev, S. Fang, C. Guo, S. He, L. Peng, S. Cao, B. Liu, L. Du, Y. Zhong and E. Polosadova, Novel Cyan-green-emitting Bi^{3+}-doped $BaScO_2F$, R^+ (R = Na, K, Rb) perovskite used for achieving full-visible-spectrum LED lighting, *Inorg. Chem.*, 2021, **60**, 15519.
99 D. Liu, X. Yun, P. Dang, H. Lian, M. Shang, G. Li and J. Lin, Yellow/orange-emitting $ABZn_2Ga_2O_7$:Bi^{3+} (A = Ca, Sr; B = Ba, Sr) phosphors: optical temperature sensing and white light-emitting diode applications, *Chem. Mater.*, 2020, **32**, 3065.
100 G. Zhou, X. Jiang, J. Zhao, M. Molokeev, Z. Lin, Q. Liu and Z. Xia, Two-dimensional-layered perovskite $ALaTa_2O_7$:Bi^{3+} (A = K and Na) phosphors with versatile structures and tunable photoluminescence, *ACS Appl. Mater. Interfaces*, 2018, **10**, 24648.

101 P. Dang, S. Liang, G. Li, Y. Wei, Z. Cheng, H. Lian, M. Shang, S. J. Ho and J. Lin, Controllable optical tuning and improvement in Li^{+}, Eu^{3+}-codoped $BaSc_2O_4$:Bi^{3+} based on energy transfer and charge compensation, *J. Mater. Chem. C*, 2018, **6**, 6449.
102 M. Noh, S.-H. Cho and S. Park, Tunable luminescence in Bi^{3+} and Eu^{3+} co-doped Sr_3AlO_4F oxyfluorides phosphors, *J. Lumin.*, 2015, **161**, 343.
103 L. I. Van Steensel, S. G. Bokhov, A. M. van de Craats, J. de Blank and G. Blasse, The luminescence of Bi^{3+} in $LaInO_3$ and some other perovskites, *Mater. Res. Bull.*, 1995, **30**, 1359.
104 T. K. V. Rao, C. S. Kamal, T. Samuel, V. S. Rao, V. V. Rao, P. V. S. S. S. N. Reddy and K. R. Rao, Color tunable luminescence from $LaAlO_3$:Bi^{3+}, Ho^{3+} doped phosphors for field emission displays, *J. Mater. Sci.: Mater. Electron.*, 2018, **29**, 1011.
105 W. Yan, Y. Wei, M. S. Molokeev, S. Wang and G. Li, Green-emitting Bi^{3+}-doped $La_2SrSc_2O_7$ phosphor for pc-WLED lighting: luminescent properties and energy transfer strategy, *J. Alloys Compd.*, 2022, **908**, 164621.
106 B. Han, J. Zhang, P. Li, J. Li, Y. Bian and H. Shi, Photoluminescence properties of novel $KBaBP_2O_8$:M (M = Pb^{2+} and Bi^{3+}) phosphors, *Opt. Mater.*, 2014, **37**, 241.
107 S. I. Kubota, H. Yamane and M. Shimada, Luminescence properties of $Gd_{1-x}Bi_xTa_7O_{19}$ ($0 < x \leq 1$), *J. Alloys Compd.*, 1998, **281**, 181.
108 S. Radhakrishna and R. S. S. Setty, Bismuth centers in alkali halides, *Phys. Rev. B: Solid State*, 1976, **14**, 969.
109 Y.-S. Chang, Blue emitting phosphors of $BaLa_2ZnO_5$ activated by bismuth ions, *J. Electrochem. Soc.*, 2001, **128**, 2027.
110 B. Jacquier, Photoluminescence processes in LaOCl:Bi and YOCl:Bi, *J. Lumin.*, 1975, **10**, 95.
111 J. G. Kang, H. M. Yoon, G. M. Chun, Y. D. Kim and T. Tsuboi, Spectroscopic studies of Bi^{3+} colour centres in KCl single crystals, *J. Phys.: Condens. Matter*, 1994, **6**, 2010.
112 A. Wolfert and G. Blasse, Luminescence of Bi^{3+} -activated LaOBr, a system with emission from different states, *J. Lumin.*, 1985, **33**, 213.
113 I. Pekgözlü, E. Erdogmus and M. Yilmaz, Synthesis and photoluminescence of $Li_4SrCa(SiO_4)_2$:M (M: Pb^{2+} and Bi^{3+}), *J. Lumin.*, 2015, **161**, 160.
114 R. Hu, Y. Zhang, Y. Zhao, X. Wang, G. Li and C. Wang, UV-Vis-NIR broadband-photostimulated luminescence of $LiTaO_3$:Bi^{3+} long-persistent phosphor and the optical storage properties, *Chem. Eng. J.*, 2020, **392**, 124807.
115 A. Wolfert and G. Blasse, Luminescence of s^2 ions in $CsCdBr_3$ and $CsMgCl_3$, *J. Solid State Chem.*, 1984, **55**, 344.
116 P. Zhiwu and S. Qiang, Luminescence of Bi^{3+} and the energy transfer from Bi^{3+} to R^{3+} (R = Eu, Dy, Sm, Tb) in alkaline-earth borates, *Solid State Commun.*, 1993, **86**, 377.
117 J. Chen, S. Zhao, Z. Zhao, M. Liao, S. Pan, J. Feng, D. Zhu, W. Pang, J. Lin and Z. Mu, The structure and luminescence properties of blue-green-emitting Sr_2YNbO_6:Bi^{3+} phosphors, *J. Lumin.*, 2021, **139**, 118336.
118 K. Polak and E. Mihokova, In^{+}, Pb^{2+} and Bi^{3+} in KBr crystals: luminescence dynamics, *Opt. Mater.*, 2010, **32**, 1208.
119 K. Li, J. Fan, M. Shang, H. Lian and J. Lin, $Sr_2Y_8(SiO_4)_6O_2$: Bi^{3+}/Eu^{3+}: a single-component white-emitting phosphor *via* energy transfer for UV w-LEDs, *J. Mater. Chem. C*, 2015, **3**, 9989.
120 S. Wang, W. Chen, D. Zhou, J. Qiu, X. Xu and X. Yu, Long persistent properties of $CaGa_2O_4$:Bi^{3+} at different ambient temperature, *J. Am. Ceram. Soc.*, 2017, **100**, 3514.
121 S. Ye, H. Liu, Y. Wang, J. Lin, K. Zhong, J. Ding and Q. Wu, Design of a bismuth-activated narrow-band cyan phosphor for use in white light emitting diodes and field emission displays, *ACS Sustainable Chem. Eng.*, 2020, **8**, 18187.
122 Z. Jiang, J. Gou, Y. Min, C. Huang, W. Lu, X. Yu, X. Su and L. Duan, Crystal structure and luminescence properties of a novel non-rare-earth activated blue-emitting garnet phosphor $Ca_4ZrGe_3O_{12}$: Bi^{3+} for n-UV pumped light-emitting diodes, *J. Alloys Compd.*, 2017, **727**, 63.
123 G. Blasse, J. de Blank and D. J. W. IJdo, The luminescence of the garnet $Ca_4ZrGe_3O_{12}$, *Mater. Res. Bull.*, 1995, **30**, 845.
124 A. M. Srivastava, Luminescence of Bi^{3+} in the orthorhombic perovskites $CaB^{4+}O_3$ (B^{4+} = Zr, Sn): crossover from localized to D-state emission, *Opt. Mater.*, 2016, **58**, 89.
125 M. Back, J. Ueda, J. Xu, K. Asami, L. Amidani, E. Trave and S. Tanabe, Uncovering the origin of the emitting states in Bi-activated $CaMO_3$ (M = Zr, Sn, Ti) perovskites: metal-to-metal charge transfer versus s-p transitions, *J. Phys. Chem. C*, 2019, **123**, 14677.
126 R. Cao, G. Quan, Z. Shi, Z. Luo, Q. Hu and S. Guo, A double perovskite Ca_2MgWO_6: Bi^{3+} yellow-emitting phosphor: synthesis and luminescence properties, *J. Lumin.*, 2017, **181**, 332.
127 R. Cao, T. Fu, H. Xu, W. Luo, D. Peng, Z. Chen and J. Fu, Synthesis and luminescence enhancement of $CaTiO_3$:Bi yellow phosphor by co-doping Al^{3+}/B^{3+} ions, *J. Alloys Compd.*, 2016, **674**, 51.
128 P. Dang, S. Liang, G. Li, Y. Wei, Z. Cheng, H. Lian, M. Shang, A. A. Al Kheraif and J. Lin, Full Color Luminescence Tuning in Bi^{3+}/Eu^{3+}-Doped $LiCa_3MgV_3O_{12}$ Garnet Phosphors Based on Local Lattice Distortion and Multiple Energy Transfers, *Inorg. Chem.*, 2018, **57**, 9251.
129 Q. Wang, Z. Mu, S. Zhang, Q. Du, Y. Qian, D. Zhu and F. Wu, Bi^{3+} and Sm^{3+} co-doped La_2MgGeO_6: a novel color-temperature indicator based on different heat quenching behavior from different luminescent centers, *J. Lumin.*, 2019, **206**, 462.
130 W. Xie, Y. W. Mo, C. W. Zou, F. W. Kang and G. H. Sun, Broad color tuning and Eu^{3+}-related photo-emission enhancement *via* controllable energy transfer in the La_2MgGeO_6:Eu^{3+}, Bi^{3+} phosphor, *Inorg. Chem. Front.*, 2018, **5**, 1076.
131 R. Wei, Z. Zheng, Y. Shi, X. Peng, H. Wang, X. Tian, F. Hu and H. Guo, Tunable emission with excellent thermal stability in single-phased SrY_2O_4:Bi^{3+}, Eu^{3+} phosphors for UV-LED, *J. Alloys Compd.*, 2018, **767**, 403.
132 M. Liao, Z. Mu, S. Zhang, F. Wu, Z. Nie, Z. Zheng, X. Feng, Q. Zhang, J. Feng and D. Zhu, A red phosphor $Mg_3Y_2Ge_3O_{12}$:Bi^{3+}, Eu^{3+} with high brightness and excellent

thermal stability of luminescence for white light-emitting diodes, *J. Lumin.*, 2019, **210**, 202.
133 S. Wang, H. Wu, Y. Fan, Q. Wang, T. Tan, R. Pang, S. Zhang, D. Li, L. Jiang, C. Li and H. Zhang, A highly efficient narrow-band blue phosphors of Bi^{3+}-activated cubic borate $Ba_3Lu_2B_6O_{15}$ towards backlight display applications, *Chem. Eng. J.*, 2022, **432**, 134265.
134 V. Babin, V. Gorbenko, A. Krasnikov, A. Makhov, M. Nikl, K. Polak, S. Zazubovich and Yu Zorenko, Peculiarities of excited state structure and photoluminescence in Bi^{3+}-doped $Lu_3Al_5O_{12}$ single-crystalline films, *J. Phys.: Condens. Matter*, 2009, **21**, 415502.
135 Y. Zorenko, J. A. Mares, R. Kucerkova, V. Gorbenko, V. Savchun, T. Voznyak, M. Nikl, A. Beitlerova and K. Jurek, Optical, luminescence and scintillation characteristics of Bi-doped LuAG and YAG single crystalline films, *J. Phys. D: Appl. Phys.*, 2009, **42**, 075501.
136 M. Nikl, A. Novoselov, E. Mihokova, K. Polak, M. Dusek, B. McClune, A. Yoshikawa and T. Fukuda, Photoluminescence of Bi^{3+} in $Y_3Ga_5O_{12}$ single-crystal host., *J. Phys.: Condens. Matter*, 2005, **17**, 3367.
137 A. M. Srivastava, M. Brik, W. W. Beers and W. E. Cohen, Chemical pressure effects in the Stokes shift of Bi^{3+} luminescence in orthorhombic perovskites, *ECS J. Solid State Sci. Technol.*, 2022, **11**, 096003.
138 A. M. Srivastava, Luminescence of Bi^{3+} in the orthorhombic perovskites $SrB^{4+}O_3$ (B^{4+} = Zr,Sn), *Opt. Mater.*, 2017, **72**, 313.
139 J. Han, F. Pan, M. S. Molokeev, J. Dai, M. Peng, W. Zhou and J. Wang, Redefinition of crystal structure and Bi^{3+} yellow luminesence with strong near-ultraviolet excitation in La_3BWO_3:Bi^{3+} phosphor for white light-emitting diodes, *ACS Appl. Mater. Interfaces*, 2018, **10**, 13660.
140 A. Wolfert and G. Blasse, Host lattice dependence of the Bi^{3+} luminescence in orthoborates $LnBO_3$, *J. Lumin.*, 1984, **31–32**, 308.
141 A. Krasnikov, L. Lipinska, E. Mihokova, M. Nikl, T. Shalapska, A. Suchocki, S. Zazubovich and Y. Zhydachevskii, Time-resolved photoluminescence and excited state structure of Bi^{3+} center in $YAlO_3$, *Opt. Mater.*, 2014, **36**, 1705.
142 W. Luo, Z. Jiang, M. Chen, Y. Dong, X. Zhu and C. Guo, Blue luminescence of Bi^{3+} in the double perovskite $CaLaMgTaO_6$ matrix for n-UV pumped white light-emitting diodes, *J. Alloys Compd.*, 2020, **814**, 152354.
143 Q. Wu, Y. Li, Y. Wang, H. Liu, S. Ye, L. Zhao, J. Ding and J. Zhou, A novel narrow-band blue-emitting phosphor of Bi^{3+}-activated $Sr_3Lu_2Ge_3O_{12}$ based on a highly symmetrical crystal structure used for WLEDs and FEDs, *Chem. Eng. J.*, 2020, **401**, 126130.
144 S. Wu, P. Xiong, X. Liu, Y. Fu, Q. Liu, Y. Chao, Q. Dong, Y. Li, W. Chen, Y. Chen, Z. Ma and M. Peng, $Sr_3Y(BO_3)_3$:Bi^{3+} phosphor with excellent thermal stability and color tunability for near-ultraviolet white-light LED, *J. Mater. Chem. C*, 2021, **9**, 3672.
145 D. Huang, P. Dang, H. Lian, Q. Zeng and J. Lin, Luminescence and energy transfer properties in Bi^{3+}/Mn^{4+}-codoped Ba_2GdNbO_6 double-perovskite phosphors for white-light-emitting diodes, *Inorg. Chem.*, 2019, **58**, 15507.
146 S. Ye, J. Ding and Q. Wu, MMCT-induced high-bright yellow light-emitting phosphor Bi^{3+}-activated Ba_2YGaO_5 used for WLED, *Chem. Eng. J.*, 2022, **428**, 13238.
147 V. Tsiumra, A. Krasnikov, S. Zazubovich, Ya Zhydachevskyy, L. Vasylechko, M. Baran, L. Wachnicki, L. Lipinska, M. Nikl and A. Suchocki, Crystal structure and luminescence studies of microcrystalline GGG:Bi^{3+} and GGG:Bi^{3+},Eu^{3+} as a UV-to-Vis converting phosphor for white LEDs, *J. Lumin.*, 2019, **213**, 278.
148 A. Krasnikov, A. Luchechko, E. Mihokova, M. Nikl, I. I. Syvorotka, S. Zazubovich and Ya Zhydachevskyy, Origin of Bi^{3+}-related luminescence in $Gd_3Ga_5O_{12}$:Bi epitaxial films, *J. Lumin.*, 2017, **190**, 81.
149 C. Li, Z. Huang, J. Wen, J. Zang, X. Tian, H. He, L. Zhang, T.-H. Huang, W. Xie and Y. Peng, Blue-emitting Bi-doped double perovskite Gd_2ZnTiO_6 phosphor with near-ultraviolet excitation for warm white light-emitting diodes, *J. Alloys Compd.*, 2019, **788**, 1127.
150 F. Kang, G. Sun, P. Boutinaud, F. Gao, Z. Wang, J. Liu, Y. Y. Li and S. Xiao, Tuning the Bi^{3+}-photoemission color over the entire visible region by manipulating secondary cations modulation in the $ScV_xP_{1-x}O_4$:Bi^{3+} ($0 \le x \le 1$) solid solution, *J. Mater. Chem. C*, 2019, **7**, 11041.
151 Y. Xue, Y. Liang, Y. Dou, H. Li, Q. Wang, X. Wu and Y. Han, Optical tuning and energy transfer of single-phase white-emitting Y_3TaO_7:Bi^{3+}, Eu^{3+} for ultraviolet converted pc-WLED with high chromatic stability, *J. Alloys Compd.*, 2023, **940**, 168917.
152 Q. Li, S. Zhang, W. Lin, W. Li, Y. Li, Z. Mu and F. Wu, A warm white emission of Bi^{3+}-Eu^{3+} and Bi^{3+}-Sm^{3+} codoping $Lu_2Ge_2O_4$ phosphors by energy transfer of Bi^{3+}-sensitized Eu^{3+}/Sm^{3+}, *Spectrochim. Acta, Part A*, 2020, **228**, 117755.
153 S.-Y. Zhu, D. Zhao, R.-J. Zhang, Y.-P. Fan and W. Liu, Utilizing the energy transfer mechanism to realize color tunable luminescence, *Dalton Trans.*, 2022, **51**, 10342.
154 F. Kang, M. Peng, S. Xu, Z. Ma, G. Dong and J. Qiu, Broadly tunable emission from $CaMoO_4$:Bi phosphor based on locally modifying the microenvironment around Bi^{3+} Ions, *Eur. J. Inorg. Chem.*, 2014, 1373.
155 Y. Zorenko, M. Pashkovsky, A. Voloshinovskii, B. Kuklinski and M. Grinberg, The luminescence of $CaWO_4$:Bi single crystals, *J. Lumin.*, 2006, **116**, 43.
156 V. Nagirnyi, A. Kotlov, L. Jönsson, M. Kirm and A. Lushchik, Emission decay kinetics in a CaWO4:Bi crystal, *Nucl. Instrum. Methods Phys. Res., Sect. A*, 2005, **537**, 61.
157 D. van der Voort and G. Blasse, Luminescence of $CaSO_4$:Bi^{3+}, a small-offset case, *J. Solid State Chem.*, 1992, **99**, 404.
158 I. Pekgözlü, E. Erdoğmuş, S. Çubul and A. S. Başak, Synthesis and photoluminescence of $LiCaBO_3$: M (M: Pb^{2+} and Bi^{3+}) phosphor, *J. Lumin.*, 2012, **132**, 1394.
159 R. Moncorgé, G. Boulon and J. P. Denis, Fluorescence properties of bismuth-doped $LaPO_4$, *J. Phys. C: Solid State Phys.*, 1979, **12**, 1165.

160 A. M. Srivastava and A. Szarowki, On the quenching of Bi^{3+} luminescence in the pyrochlore Gd_2GaSbO_7, *J. Solid State Chem.*, 1999, **146**, 494.

161 Y. Wang, S. Yang, Q. Xu, C. Yang, S. Zhang, Y. Yu and M. Jiao, Luminescence property, color tunability, and optical thermometry behavior of bismuth-europium codoped strontium-anorthite-type phosphors, *J. Phys. Chem. C*, 2022, **126**(36), 15517.

162 D. R. Taikar, S. Tamboli and S. J. Dhoble, Synthesis and photoluminescence properties of red, green and blue emitting $LaYO_3$:M (M = Eu^{3+}, Tb^{3+}, Sm^{3+}, Bi^{3+}, Pb^{2+}) phosphors, *Optik*, 2017, **142**, 183.

163 D. Huang, Y. Wei, P. Dang, X. Xiao, H. Lian and J. Lin, Tunable color emission in $LaScO_3$:Bi^{3+}, Tb^{3+}, Eu^3+ phosphor, *J. Am. Ceram. Soc.*, 2020, **103**, 3273.

164 H. Yang, P. Li, X. Fu, Z. Ye, X. Huo, Y. Wang, Q. Wu, H. Suo, L. Li and Z. Wang, A novel blue emitting phosphor $LaScO_3$:Bi^{3+} for white LEDs: high quantum efficiency, high color purity and excellent thermal stability, *Mater. Today Chem.*, 2022, **26**, 101050.

165 J. Wang, R. Lei, S. Zhao, F. Huang, D. Deng, S. Xu and H. Wang, Color tunable Bi^{3+}/Eu^{3+} co-doped La_2ZnTiO_6 double perovskite phosphor for dual-mode ratiometric optical thermometry, *J. Alloys Compd.*, 2021, **881**, 160601.

166 A. Madej, M. E. Witkowski, A. J. Wojtowicz and E. Zych, Photo- and radioluminescent properties of undoped and Bi-doped Lu_2WO_6 powders at 10–300 K, *J. Lumin.*, 2015, **160**, 50.

167 A. M. Srivastava, H. A. Comanzo and M. G. Brik, Luminescence of Bi^{3+} in the double perovskite La_2MgTiO_6, *Opt. Mater.*, 2018, **75**, 809.

168 A. M. Srivastava, H. A. Comanzo and M. G. Brik, The nature of Bi^{3+} luminescence in the double perovskite La_2LiSbO_6, *J. Lumin.*, 2017, **192**, 620.

169 M. Song, W. Ran, Y. Ren, L. Wang and W. Zhao, Characterizations and photoluminescence properties of a dual-functional La_2LiNbO_6:Bi^{3+}, Eu^{3+} phosphor for WLEDs and ratiometric temperature sensing, *J. Alloys Compd.*, 2021, **865**, 158825.

170 F. Kang, M. Peng, Q. Zhang and J. Qiu, Abnormal anti-quenching and controllable multi-transitions of Bi^{3+} luminescence by temperature in a yellow-emitting $LuVO_4$:Bi^{3+} phosphor for UV-converted white LEDs, *Chem. – Eur. J.*, 2014, **20**, 11522.

171 P. Guo, F. Zhao, G. Li, F. Liao, S. Tian and X. Jing, Novel phosphors of Eu^{3+}, Tb^{3+} or Bi^{3+} activated Gd_2GeO_5, *J. Lumin.*, 2003, **105**, 61.

172 T. K. Park, H. C. Ahn and S. I. Mho, High Concentration of Bi^{3+} Incorporated into $RNbO_4$:Eu^{3+} (R = La, Y, Gd) as Red Phosphors for White LED Applications, *J. Korean Phys. Soc.*, 2008, **52**, 431.

173 S. H. Shin, D. Y. Jeon and K. S. Suh, Charge-transfer nature in luminescence of $YNbO_4$:Bi blue phosphor, *J. Appl. Phys.*, 2001, **90**, 5986.

174 Q. Wu, X. Chen, H. Chen, H. Zhang and J. Zhou, A novel pale-yellow $Ba_2ZnGe_2O_7$:Bi^{3+} phosphor with site-selected excitation and small thermal quenching, *J. Am. Ceram. Soc.*, 2019, **102**, 6068.

175 G. Boulon, Processus de photoluminescence dans les oxydes et les orthovanadates de terres rares polycristallins activés par l'ion Bi^{3+}, *J. Phys.*, 1971, **32**, 333.

176 E. Cavalli, F. Angiuli, F. Mezzadri, M. Trevisani, M. Bettinelli, P. Boutinaud and M. G. Brik, Tunable luminescence of Bi^{3+}-doped $YP_xV_{1-x}O_4$ ($0 \leq x \leq 1$), *J. Phys.: Condens. Matter*, 2014, **26**, 385503.

177 G. Blasse and A. Bril, Luminescence phenomena in compounds with fergusonite structure, *J. Lumin.*, 1970, **3**, 109.

178 Y. Katayama, J. Ueda and S. Tanabe, Effect of Bi_2O_3 doping on persistent luminescence of $MgGeO_3$:Mn^{2+} phosphor, *Opt. Mater. Express*, 2014, **4**, 613.

179 J. Li, S. Zhang, H. Luo, Z. Mu, Z. Li, Q. Du, J. Feng and F. Wu, Efficient near ultraviolet to near infrared down-conversion photoluminescence of La_2GeO_5:Bi^{3+}, Nd^{3+} phosphor for silicon-based solar cells, *Opt. Mater.*, 2018, **85**, 523.

180 K. Liu, C. Si, L. Lv, H. Li and J. Zhang, An excitation wavelength-dependent color-tunable La_4GeO_8:Bi^{3+}, Er^{3+} phosphor for multicolor anticounterfeiting, *J. Alloys Compd.*, 2022, **910**, 164836.

181 B. N. Mahalley, S. J. Dhoble, R. B. Pode and G. Alexander, Photoluminescence in $GdVO_4$:Bi^{3+},Eu^{3+} red phosphor, *Appl. Phys. A: Mater. Sci. Process.*, 2000, **70**, 39.

182 Y. Zorenko, V. Gorbenko, T. Zorenko, P. Malinowski, V. Jary, R. Kucerkova, A. Beitlerova, J. A. Mares, M. Nikl and A. Fedorov, Luminescent and scintillation properties of Bi^{3+} doped Y_2SiO_5 and Lu_2SiO_5 single crystalline films, *J. Lumin.*, 2014, **154**, 525.

183 V. Gorbenko, A. Krasnikov, E. Mihokova, M. Nikl, S. Zazubovich and Y. Zorenko, Photoluminescence and excited state structure of Bi^{3+}- related centers in Lu_2SiO_5:Bi single crystal films, *J. Lumin.*, 2013, **134**, 469.

184 S. F. Wang, F. Gu, M. K. Lü, W. G. Zou, S. W. Liu, D. Xu, D. R. Yuan and G. J. Zhou, Photoluminescence characteristics of $ZnTiO_3$:Bi^{3+} nanocrystals, *J. Phys. Chem. Solids*, 2004, **65**, 1243.

185 V. Babin, V. Gorbenko, A. Krasnikov, E. Mihokova, M. Nikl, S. Zazubovich and Y. Zorenko, Photoluminescence and excited state structure in Bi^{3+}-doped Y_2SiO_5 single crystalline films, *Radiat. Meas.*, 2013, **56**, 90.

186 H. S. Zhang, F. W. Kang, Y. J. Zhao, M. Y. Peng, D. Y. Lei and X. B. Yang, The role of oxygen defects in a bismuth doped $ScVO_4$ matrix: tuning luminescence by hydrogen treatment, *J. Mater. Chem. C*, 2017, **3**, 314.

187 P. Boutinaud, A. Barros and F. Kang, Luminescence depreciation in $ScVO_4$:Bi^{3+} upon irradiation in the Bi^{3+}-related absorption bands, *J. Lumin.*, 2022, **248**, 118941.

188 V. G. Suchithra, P. P. Rao and B. A. Aswathy, Color-tunable phosphors in weberite type system, La_3SbO_7:Bi^{3+}, Eu^{3+} for near-UV LED application, *ChemistrySelect*, 2017, **2**, 7602.

189 Z. Tan, J. Li, C. Zhang, Z. Li, Q. Hu, Z. Xiao, T. Kamiya, H. Hosono, G. Niu, E. Lifshitz, Y. Cheng and J. Tang, Highly efficient blue-emitting Bi-doped Cs_2SnCl_6

perovskite variant: photoluminescence induced by impurity doping, *Adv. Funct. Mater.*, 2018, **28**, 1801131.
190 Y. Zhuang, J. Ueda and S. Tanabe, Photochromism and white ion-lasting persistent luminescence in Bi^{3+}-doped $ZnGa_2O_4$ ceramics, *Opt. Express*, 2012, **2**, 1378.
191 A. C. van der Steen, Luminescence of Cs_2NaYCl_6: Bi^{3+} ($6s^2$), *Phys. Status Solidi B*, 1980, **100**, 603.
192 G. Blasse and G. J. Dirksen, The luminescence of broad-band emitters in $LiLaP_4O_{12}$, *Phys. Status Solidi B*, 1982, **110**, 487.
193 A. C. van der Steen, J. J. A. van Hesteren and A. P. Slok, Luminescence of Bi^{3+} ion in $LiLnO_2$ and $NaLnO_2$ (Ln = Sc, Y, La, Gd, Lu), *J. Electrochem. Soc.*, 1981, **128**, 1327.
194 A. F. Ellervee, Luminescence of Pb^{2+} and Bi^{3+} centres in alkali-earth sulphides and oxides, *Phys. Status Solidi B*, 1977, **82**, 91.
195 N. Yamashita and S. Asano, Luminescence centers of Ca(SSe):Bi^{3+} and CaO: Bi^{3+} phosphors, *J. Phys. Soc. Jpn.*, 1976, **40**, 144.
196 A. E. Hughes and G. P. Pells, The luminescence spectra of Bi^{3+} ions in MgO and CaO, *Phys. Status Solidi B*, 1975, **71**, 707.
197 R. Aceves, M. Barboza Flores, A. Maaroos, V. Nagirnyi, R. Perez Salas, T. Tsuboi, S. Zazubovich and V. Zepelin, RES structure of Bi^{3+} centres in KCl:Bi, S and CaO:Bi crystals, *Phys. Status Solidi B*, 1996, **194**, 619.
198 V. P. Dotsenko, N. P. Efryushina and I. B. Berezovskaya, Luminescence properties of $GaBO_3$: Bi^{3+}, *Mater. Lett.*, 1996, **28**, 517.
199 F. Pelle, B. Jacquier, J. P. Denis and B. Blanzat, Optical properties of $Cs_2NaBiCl_6$, *J. Lumin.*, 1978, **17**, 61.
200 G. Dirksen and G. Blasse, Tetracalcium gadolinium oxoborate ($Ca_4GdO(BO_3)_3$) as a new host lattice for luminescent materials, *J. Alloys Compd.*, 1993, **191**, 121.
201 Y. Zhang, D. Chen, W. Wang, S. Yan, J. Liu and Y. Liang, Long-lasting ultraviolet-A persistent luminescence and photostimulated persistent luminescence in Bi^{3+}-doped $LiScGeO_4$ phosphor, *Inorg. Chem. Front.*, 2020, **7**, 3063.
202 L. Chen, Y. Long, Y. Qin and W. Li, Co-precipitation preparation, characterization and optical properties of blue $CaSb_2O_6$:Bi^{3+} nano-phosphor, *Mater. Lett.*, 2013, **102–103**, 59.
203 G. Ju, Y. Hu, L. Chen, X. Wang, Z. Mu, H. Wu and F. Kang, The luminescence of bismuth and europium in $Ca_4YO(BO_3)_3$, *J. Lumin.*, 2012, **132**, 717.
204 B. Jacquier, G. Boulon, G. Sallavuard and F. Gaume-Mahn, Bi^{3+} center in a lanthanum gallate phosphor, *J. Solid State Chem.*, 1972, **4**, 374.
205 A. M. Srivastava, Luminescence of Bi^{3+} in $LaGaO_3$, *Mater. Res. Bull.*, 1999, **34**, 1391.
206 O. M. Bordun, Luminescence of bismuth-activated ceramics of yttrium and scandium oxides, *J. Appl. Spectrosc.*, 2002, **69**, 67.
207 G. Boulon, C. Pedrini, M. Guidoni and C. Pannel, Étude de la cinétique des centres luminogènes Bi^{3+} dans les cristaux, *J. Phys.*, 1975, **36**, 267.
208 Y. Fu, P. Xiong, X. Liu, X. Wang, S. Wu, Q. Liu, M. Peng and Y. Chen, A promising blue-emitting phosphor $CaYGaO_4$:Bi^{3+} for near-ultraviolet (NUV) pumped white LED application and the emission improvement by Li^+ ions, *J. Mater. Chem. C*, 2021, **9**, 303.
209 A. Wolfert and G. Blasse, Luminescence of Bi^{3+}-doped crystals of Cs_2NaYBr_6 and $Cs_2NaLaCl_6$, *J. Solid State Chem.*, 1985, **59**, 133.
210 T. Wang, X. Xu, D. Zhou, Y. Yang, J. Qiu and X. Yu, Effect of defect distribution on the low optical storage properties of strontium gallates with a low-dimensional chain structure, *Inorg. Chem.*, 2016, **55**, 894.
211 N. Yamashita, S. Ikeda and S. Asano, Photoluminescence and excitation spectra of the SrO:Bi^{3+} phosphor, *Phys. Lett. A*, 1987, **121**, 94.
212 A. Yousif, R. M. Jafer, S. Som and H. C. Swart, The effect of the host lattice on the optical properties of Bi^{3+} in $Ca_{1-x}O$:Bi and $Ca_{1-x}(OH)_2$:Bi phosphors, *Appl. Surf. Sci.*, 2018, **433**, 155.
213 H. Wang, X. Chen, L. Teng, D. Xu, W. Chen, R. Wei, F. Hu, X. Sun and H. Guo, Adjustable emission and energy transfer process in $BaGd_2O_4$:Bi^{3+}, Eu^{3+} phosphors, *J. Lumin.*, 2019, **206**, 185.
214 H. Cai, Z. Song and Q. Liu, Infrared-photostimulable and long-persistent ultraviolet-emitting phosphor $LiLuGeO_4$:Bi^{3+}, Yb^{3+} for biophotonic applications, *Mater. Chem. Front.*, 2021, **5**, 1468.
215 L. Sun, B. Devakumar, J. Liang, S. Wang, Q. Sun and X. Huang, Novel high-efficiency violet-red dual-emitting Lu_2GeO_5:Bi^{3+}, Eu^{3+} phosphors for indoor plant growth lighting, *J. Lumin.*, 2019, **214**, 116544.
216 S. Asano and N. Yamashita, Luminescence et interaction phonon-electron dansle luminophore MgS:Bi^{3+}, *Phys. Status Solidi B*, 1981, **105**, 305.
217 W. Zhao, S. Qi, J. Liu and B. Fan, Luminescence properties and energy transfer of a color-tunable Y_2GeO_5:Bi^{3+}, Eu^{3+} phosphor, *J. Alloys Compd.*, 2019, **787**, 469.
218 B. M. Jafer, H. C. Swart, H. A. A. Seed Ahmed, A. Yousif and R. E. Kroon, Optical properties and stability of Bi doped La_2O_2S, *Opt. Mater.*, 2019, **95**, 109260.
219 F. Kellendonk, T. van den Belt and G. Blasse, On the luminescence of bismuth, cerium and chromium and yttrium aluminium borate, *J. Chem. Phys.*, 1982, **76**, 1194.
220 Y. Zhydachevskyy, V. Tsiumra, M. Baran, L. Lipinska, P. Sybilski and A. Suchocki, Quantum efficency of the down-conversion process in Bi^{3+}–Yb^{3+} co-doped Gd_2O_3, *J. Lumin.*, 2018, **196**, 169.
221 G. X. Liu, R. Zhang, Q. L. Xiao, S. Y. Zou, W. F. Peng, L. W. Cao and J. X. Meng, Efficient $Bi^{3+} \rightarrow Nd^{3+}$ energy transfer in Gd_2O_3: Bi^{3+}, Nd^{3+}, *Opt. Mater.*, 2011, **34**, 313.
222 Y.-L. Yang, Y. Zhou, D.-J. Pan, Z.-J. Zhang and J.-T. Zhao, The luminescence properties of CaZnOS: Bi^{3+}, Sm^{3+}, Li^+ phosphors with tunable emissions and energy transfer for white emission, *J. Lumin.*, 2019, **206**, 578.
223 L. Liu, K. Yu, L. Ming, Y. Sheng, S. Zheng, L. Song, J. Shi and Y. Zhang, A novel Gd-based phosphor $NaGdGeO_4$:Bi^{3+},

Li^{+} with super-long ultraviolet-A persistent luminescence, *J. Rare Earths*, 2022, **40**, 1424.

224 N. Yamashita, S. Iwasaki, S. Asano, M. Ohishi and K. Ohmori, Investigation of the luminescence in the $SrS:Bi^{3+}$ phosphor by time-resolved emission spectroscopy, *J. Phys. Soc. Jpn.*, 1984, **53**, 4425.

225 S. Asano, N. Nakao, N. Yamashita and I. Matsuyama, Luminescence et transition non radiative $^3T_{1u} \rightarrow {}^3A_{1u}$ dans le luminophore $BaS:Bi^{3+}$, *Phys. Status Solidi B*, 1986, **133**, 333.

226 P. Boutinaud, Rationalization of the Pr^{3+}-to-transition metal charge transfer model: Application to the luminescence of Pr^{3+} in titano-niobates, *J. Lumin.*, 2019, **214**, 116557.

227 R. H. P. Awater, L. C. Niemeijer-Berghuijs and P. Dorenbos, Luminescence and charge carrier trapping in YPO_4: Bi, *Opt. Mater.*, 2017, **66**, 351.

228 A. Jain, G. Hautier, C. Moore, S. P. Ong, C. C. Fischer, T. Mueller, K. A. Persson, G. Ceder and A. High-Throughput, Infrastructure for Density Functional Theory Calculations, *Comput. Mater. Sci.*, 2011, **50**, 2295.

229 M. Colmont, P. Boutinaud, C. Latouche, F. Massuyeau, M. Huvé, A. Zadoya and S. Jobic, Origin of luminescence in La_2MoO_6 and $La_2Mo_2O_9$ and their Bi-doped variants, *Inorg. Chem.*, 2020, **59**, 3215.

230 F. Du, Z. Tang, Q. Zhao, L. Du and W. Xia, $Ba_5GeO_4Br_6:Bi^{3+}$, a promising cyan phosphor for high-quality full-spectrum white light illumination, *J. Lumin.*, 2023, **255**, 119592.

231 P. He, W. Ge, Y. Wang, Y. Tian and Q. Zhang, Multicolor-emission of $Gd_2O_3:Bi^{3+}$ and its application for white light emitting diodes, *Optik*, 2023, **274**, 170523.

232 H. Kong, G. Jia, H. Li, Z. Meng, N. Zhang and C. Zhang, Deep blue, cyan, orange-red, and white multicolor emissions generated by Bi^{3+}/Eu^{3+} activated $KBaYSi_2O_7$ luminescent materials for white light-emitting diodes, *Ceram. Interfaces*, 2023, DOI: 10.1016/j.ceramint.2023.01.116.

233 X. Wu, X. Zhao, Q. Ren, L. Du, M. Pei and O. Hai, Design efficient energy transfer $Ca_2Al_2SiO_7$: Bi^{3+}, Eu^{3+} phosphors by cationic substitution for full-spectrum W-LED lighting, *Ceram. Interfaces*, 2023, DOI: 10.1016/j.ceramint.2023.03.007.

234 R. Liu, H. Wu, S. Wang, W. Yuan, S. Zhang, R. Pang, L. Jiang, D. Li, C. Li and H. Zhang, $Ba_3LuGa_2O_{7.5}$: Bi^{3+} phosphors with potential application in full-spectrum WLEDs and temperature sensing, *J. Mater. Chem. C*, 2023, **11**, 2653.

235 H. Guo, Y. Chen, S. Gu, L. Wang, Q. Shi, L. Zhao, C. Cui, Y. Cui and P. Huang, UV-light activated blue-emitting persistent luminescence of $CaYGaO_4:Bi^{3+}$ suitable for high-temperature environment, *J. Lumin.*, 2023, **255**, 119602.

236 M. Baran, A. Kissabekova, A. Krasnikov, V. Tsiumra, L. Vasylechko, S. Zazubovich and Ya Zhydachevskyy, Bi^{3+}–doped garnets as possible ultraviolet persistent phosphors, *Opt. Mater.*, 2023, **137**, 113584.

237 C. Wang, Y. Jin, J. Zhang, X. Li, H. Wu, R. Zhang, Q. Yao and Y. Hu, Linear charging-discharging of an ultralong UVA persistent phosphor for advanced optical data storage and wide-wavelength-range detector, *Chem. Eng. J.*, 2023, **453**, 139558.

238 W. Yuan, R. Pang, S. Wang, T. Tan, J. Su, X. Chen, R. Liu, C. Li, S. Zhang and H. Zhang, Blue-light activated green emitting $BaY_2Al_2Ga_2SiO_{12}:Ce^{3+},Bi^{3+}$: efficient persistent luminescence phosphors for AC-LEDs, *Inorg. Chem. Front.*, 2023, **10**, 1203.

239 M. Cai, T. Lang, S. Fang, T. Han, D. Valiev, H. You, C. Liu, J. Yu, P. Su, X. Jing, G. Ge, B. Liu and E. F. Polisadova, Color tunable $(Ba,Ca)ScO_2F$: Eu^{2+}, Bi^{3+}, K^{+} perovskite with dependence of excitation wavelength for advanced anti-counterfeiting application, *J. Lumin.*, 2023, **257**, 119713.

240 Q. Chen, H. Fu, J. Jiang, Z. Fang, H. Zhang, W. Yang, W. Liu and J. Zheng, Zr^{4+} and Bi^{3+} codoped $Cs_2Ag_{0.3}Na_{0.7}InCl_6$ double perovskite for single-composition white-light emitting phosphors and multimodal optical anti-counterfeiting, *J. Lumin.*, 2023, **258**, 119783.

241 H. Zhang, Y. Wang, Y. Jiao, S. Chen, M. Liao, Y. Fang, W. Gao and L. Hu, Excitation-wavelength-dependent photochromic Bi^{3+}/Ln^{3+} (Eu^{3+}, Er^{3+})-codoped phosphors for anticounterfeiting applications, *Ceram. Interfaces*, 2023, DOI: 10.1016/j.ceramint.2023.02.156.

242 Y. Li, S. Xu, Q. Zhang, Y. Gao, X. Zhang, Y. Cao, H. Yu, X. Li and B. Chen, Excitation wavelength-dependent multi-emission in $Sb^{3+}/Bi^{3+}/Er^{3+}$ codoped perovskite toward optical anti-counterfeiting and information storage, *J. Alloys Compd.*, 2023, **940**, 168925.

243 S. Xu, J. Lei, L. Li, J. Chen, L. Chen and H. Guo, Dual-mode optical thermometry of $Sr_2YNbO_6:Bi^{3+},Eu^{3+}$ phosphors designed by response surface methodology, *J. Lumin.*, 2023, **255**, 119615.

244 Y. Hu, X. Li, K. Wang, Z. Guan, H. Yu, Y. Zhang, S. Xu and B. Chen, Tunable photoluminescence properties and non-contact thermometry of Bi^{3+}, Sm^{3+} co-doped YVO_4 phosphors, *J. Lumin.*, 2023, **257**, 119722.

245 Y. Luo, D. Zhang, S. Xu, L. Li, L. Chen and H. Guo, Optical thermometry based on Bi^{3+},Ln^{3+} co-doped $YNbO_4$ (Ln = Dy, Eu) phosphors, *J. Lumin.*, 2023, **257**, 119780.

246 B. Zhao, Y. Chen, Y. Xue, Q. Mao, G. Bai, M. Liu and J. Zhong, Tunable emission color and anti-thermal-quenching behaviors in niobates for high-sensitive optical thermometry, *Mater. Des.*, 2023, **227**, 111802.

247 L. Ma, F. Lu, Q. Yu, P. Dai, F. Hu, H. Guo and R. Wei, A three-mode optical thermometry based on thermochromic Gd_2GaSbO_7: Bi^{3+}, Eu^{3+} phosphors, *Ceram. Interfaces*, 2023, DOI: 10.1016/j.ceramint.2023.02.029.

248 Z. Zhang, X. Yu, X. Xue, B. Gao, Y. Bai, H. Yang, X. Li, J. Fan and P. Zhao, Enhanced upconversion luminescence and temperature sensing performance in Er^{3+}/Yb^{3+}-codoped K_3ScF_6 phosphors induced by tridoping with Bi^{3+} ions, *J. Alloys Compd.*, 2023, **930**, 167357.

249 R. Song, S. Xu, Y. Li, Q. Zhang, Y. Gao, H. Yu, Y. Cao, X. Li, S. Zhang and B. Chen, Bi^{3+} and Tb^{3+} Co-doped $Cs_2AgInCl_6$ Lead-free double perovskite nanocrystals for detection of temperature and copper ions, *Spectrochim. Acta, Part A*, 2023, **288**, 122181.

